



Chapter Two

Green's functions

in quantum transport

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Course lecture notes

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Equilibrium correlations and spectral functions; equations of motion; the closed time contour; contact transport and connected current noise.

September 2026

English study and teaching edition

Learning route

The chapter develops one continuous question: how does the quantum motion of fermions become a measurable current and its fluctuations? A two-time correlation first records the amplitude and occupation information that a single expectation value cannot carry. Causal Green's functions then let us solve dynamics, while the lesser and greater components retain how the available states are filled. The closed time contour combines these roles for a specified initial preparation. Its component algebra finally reduces a reservoir contact to transmission, current, and noise.

The derivations assume basic operator quantum mechanics. The needed trace identities, canonical anticommutation relations, Fourier measures, contour signs, and matrix operations are introduced where they enter. Throughout, $\hbar = k_B = 1$ unless units are explicitly restored, $e > 0$ is the elementary charge, and $h = 2\pi\hbar$. Each reservoir label denotes one independent fermionic channel unless an additional sum is stated.

The reading references open the chapter. The physical discussion then follows the progression of Chapter Two, from equilibrium to the two complementary noise calculations. A symbol table at the end provides a compact route back to the definitions.

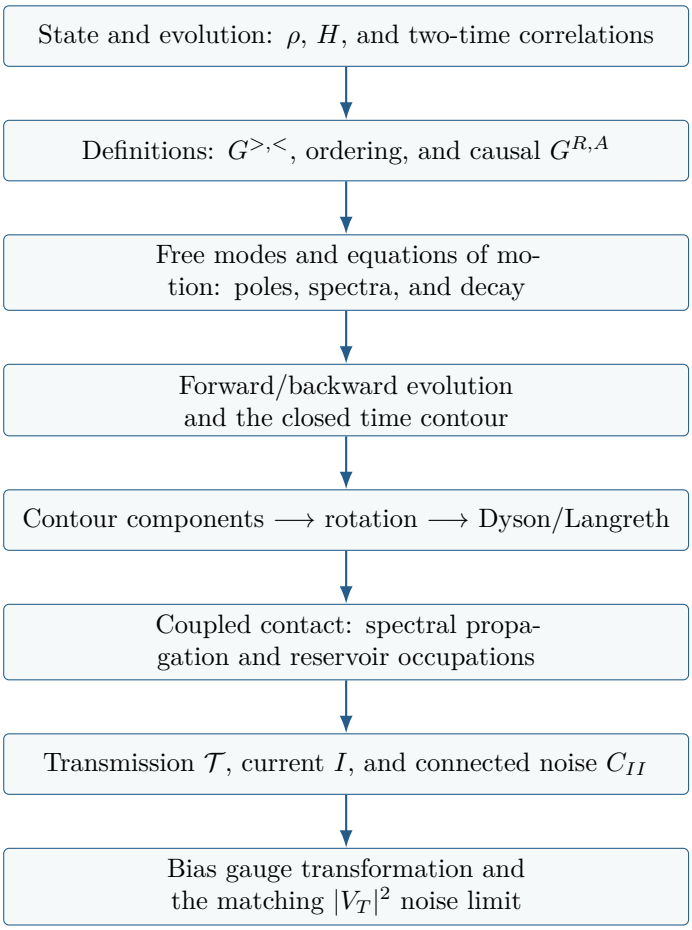


Figure 2.1: Conceptual dependencies in this chapter (schematic). The vertical path follows the calculation from a prepared state to transport observables. Spectral information and occupation information remain distinct until they enter the contact calculation together. The final gauge calculation tests the weak-contact limit of the dressed calculation.

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References for the chapter

For the basic ideas, consult Joseph Maciejko's 2007 notes, *An Introduction to Nonequilibrium Many-Body Theory* [1]. For free-lead Green's functions and the equation-of-motion method, the reference is Henrik Bruus and Karsten Flensberg, *Many-Body Quantum Theory in Condensed Matter Physics: An Introduction* [2].

For an extension to full counting statistics, the reference sheet points to A. O. Gogolin and A. Komnik, *Physical Review B* **73**, 195301 (2006), "Towards full counting statistics for the Anderson impurity model" [3]. The present chapter develops the current and its second connected correlation. The last reference identifies a further direction; the derivations below concern the Green's-function contact problem itself.

Chapter Two: Green's functions in quantum transport.

Green's functions organize two complementary questions. Which excitations can a system support, and how are those excitations populated? When systems are connected, the same two-point functions also provide the building blocks for current and current fluctuations. Keeping spectral information and occupation information distinct will be essential throughout this chapter.

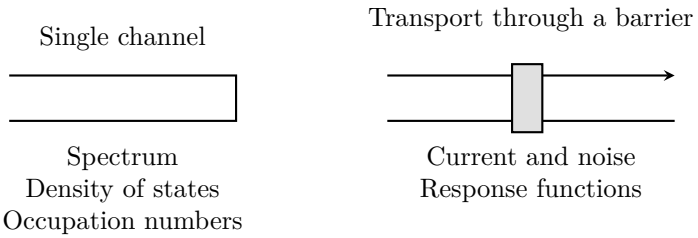


Figure 2.2: The two starting settings of the lecture. A single channel motivates the study of energy levels, density of states, and occupations. A channel containing a barrier motivates transport observables: current, noise, and response functions. The arrow in the transport sketch indicates a chosen propagation direction; no numerical transmission is implied.

For a single channel, we seek its *energy spectrum*; its *density of states*; and its *occupation numbers*. We set $\hbar = 1$ for simplicity. We also set $k_B = 1$, so that temperature has energy units and $\beta = 1/T$. When units are restored later, $h = 2\pi\hbar$.

2.1 Review of equilibrium Green's functions and the equation of motion

2.1.1 Basic concepts: equilibrium states

Consider a time-independent, number-conserving Hamiltonian H on the many-body Hilbert space, with total number operator N and $[H, N] = 0$. The density matrix specifies statistical weights of states; it is a different object from the Hamiltonian that generates real-time motion. In the grand-canonical ensemble, the system exchanges energy and particles with an equilibrium reservoir. The equilibrium density matrix is

$$\rho_{\mathrm{eq}} = \frac{1}{Z} e^{-\beta(H-\mu N)}. \tag{2.1.1}$$

The normalization is the grand partition function,

$$Z = \mathrm{Tr}\, e^{-\beta(H-\mu N)}. \tag{2.1.2}$$

Here H is the Hamiltonian, N counts particles, μ is the chemical potential, and $\beta = 1/T$ is the inverse temperature. The state has the equilibrium distribution in (2.1.1); the equilibrium model considered here has $H(t) = H$. It also satisfies

$$[\rho_{\mathrm{eq}}, H] = 0. \tag{2.1.3}$$

The latter follows because $[H, H-\mu N]=0$: every power in the exponential commutes with H. It is useful to separate the two claims. The Gibbs form fixes thermal weights; commutation with H fixes stationarity. An arbitrary density matrix diagonal in the energy basis is stationary, but its weights need not be thermal. Likewise, a time-independent Hamiltonian alone does not force an arbitrary initial state into equilibrium.

The expectation value of an operator O is

$$\langle O\rangle = \mathrm{Tr}\left(O\frac{1}{Z}e^{-\beta(H-\mu N)}\right) \tag{2.1.4}$$

$$= \mathrm{Tr}(\rho_{\mathrm{eq}}O). \tag{2.1.5}$$

The second equality uses cyclicity of the trace. For simultaneous eigenstates $|m\rangle$ of H and N, this means

$$\langle O\rangle = \sum_m p_m\langle m|O|m\rangle, \qquad p_m = Z^{-1}e^{-\beta(E_m-\mu N_m)}. \tag{2.1.6}$$

Thus an expectation combines quantum matrix elements with statistical weights.

2.1.2 Two-time correlations and time-translation invariance

A Green's function is built from correlations between operators at different times. For two general operators A and B, begin with the raw correlator

$$C_{AB}(t,t') = \langle A(t)B(t')\rangle = \mathrm{Tr}[\rho_{\mathrm{eq}}A(t)B(t')]. \tag{2.1.7}$$

The operators are in the Heisenberg picture:

$$A(t) = e^{iHt}Ae^{-iHt}, \tag{2.1.8}$$

$$B(t') = e^{iHt'}Be^{-iHt'}. \tag{2.1.9}$$

The reference time is zero. Neither operator has explicit time dependence in the Schrödinger picture. Here real-time evolution uses H, whereas the thermal weights use $\mathcal{K}=H-\mu N$. These generators must not be interchanged without an energy-reference transformation.

Since $[\rho_{\mathrm{eq}}, H]=0$, the correlator depends only on the elapsed time. Retaining the explicit trace manipulations,

$$C_{AB}(t,t') = \mathrm{Tr}\left[\rho_{\mathrm{eq}}e^{iHt}Ae^{-iH(t-t')}Be^{-iHt'}\right]. \tag{2.1.10}$$

Move the rightmost exponential to the beginning of the trace, commute it past ρ_{eq}, and combine exponentials of the same Hamiltonian:

$$C_{AB}(t,t') = \mathrm{Tr}\left[e^{-iHt'}\rho_{\mathrm{eq}}e^{iHt}Ae^{-iH(t-t')}B\right] \tag{2.1.11}$$

$$= \mathrm{Tr}\left[\rho_{\mathrm{eq}}e^{-iHt'}e^{iHt}Ae^{-iH(t-t')}B\right]. \tag{2.1.12}$$

$$C_{AB}(t,t') = \mathrm{Tr}\left[\rho_{\mathrm{eq}}e^{iH(t-t')}Ae^{-iH(t-t')}B\right] = \langle A(t-t')B(0)\rangle \tag{2.1.13}$$

$$\equiv C_{AB}(t-t'). \tag{2.1.14}$$

This proof uses trace cyclicity and a stationary state. It does not require the two operators to commute, and it does not permit their order to be reversed. In particular, $C_{AB}(t-t')$ and $C_{BA}(t'-t)$ generally differ. The same stationarity argument applies to a stationary nonequilibrium state when the necessary commutation condition and trace are well defined; thermal relations impose stronger conditions. For driven transients, retain both arguments t, t'.

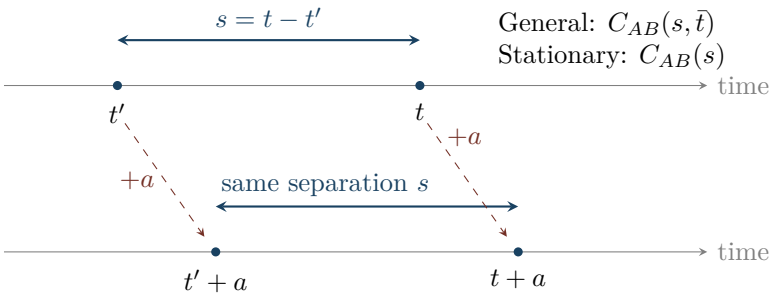


Figure 2.3: Two times and their common translation. The separation $s = t - t'$ is unchanged when both times are shifted by a. For a stationary state evolving with a time-independent Hamiltonian, $C_{AB}(t+a, t'+a) = C_{AB}(t, t') = C_{AB}(s)$. A general transient may additionally depend on the mean time $\bar{t} = (t + t')/2$, which increases by a under this translation.

2.1.3 Definitions of different Green's functions

The definitions below apply also out of equilibrium, although equilibrium simplifications may no longer apply. Start with one canonical fermionic mode, with

$$\{c^\dagger, c\} = 1. \tag{2.1.15}$$

For several modes the algebra is $\{c_i, c_j^\dagger\} = \delta_{ij}$ and $\{c_i, c_j\} = 0$. The first Green's-function index belongs to the annihilation operator: G_{ij} contains c_i and $c_j^\dagger$. Until stationarity has been established, both time arguments must be displayed. In the stationary equal-mode formulas below, the final single-argument expression is shorthand for the same two-time function.

The *greater* Green's function is

$$G^>(t, t') = -i\langle c(t)c^\dagger(t')\rangle \underset{\text{stationary}}{=} G^>(t - t'). \tag{2.1.16}$$

The *lesser* Green's function is

$$G^<(t, t') = +i\langle c^\dagger(t')c(t)\rangle \underset{\text{stationary}}{=} G^<(t - t'). \tag{2.1.17}$$

Both expressions pair the same operator times, t and t', but reverse the operator order inside the expectation value. The correspondence tracks the operator arguments and retains the same state.

The greater correlator describes creating a particle at time t' and annihilating it at time t. Relative to a zero-temperature Fermi sea, this is particle propagation above the filled sea. The lesser correlator instead describes the hole sector associated with occupied states below the Fermi surface. These particle and hole pictures refer to matrix elements and the reference state. For $t > t'$, the greater product acts on a ket from right to left: $c^\dagger(t')$ adds a particle and $c(t)$ removes it. For the lesser product, one may view the missing-particle amplitude by following the reverse ordering. For a single free mode the weights are $1-f$ and f, respectively. At nonzero temperature they are smoothly distributed across the chemical potential; "above" and "below" are then a useful zero-temperature limit rather than strict support conditions. In an interacting system, addition and removal can excite many many-body eigenstates rather than a single sharp level.

The lesser function also contains the occupation number. At equal times the relation is especially direct:

$$n(t) = \langle c^\dagger(t)c(t)\rangle = -iG^<(t, t). \tag{2.1.18}$$

For a stationary function, fix the Fourier pair now:

$$G(\omega) = \int_{-\infty}^{\infty} dt\, e^{i\omega t}G(t), \qquad G(t) = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} e^{-i\omega t}G(\omega). \tag{2.1.19}$$

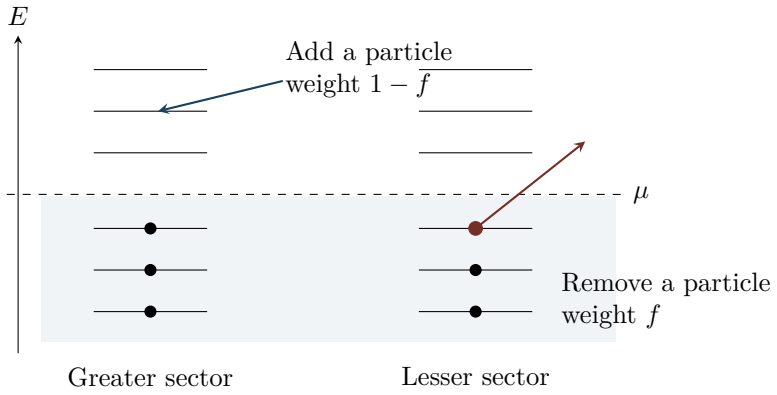


Figure 2.4: Addition and removal relative to a zero-temperature Fermi sea. An empty level carries addition weight $1-f$, whereas an occupied level carries removal weight f. This schematic labels spectral sectors; it does not identify an interacting excitation with a bare free-particle state.

The Fourier transform of the lesser function gives energy-resolved occupied spectral weight:

$$W_{\mathrm{occ}}(\omega) \equiv -i\int_{-\infty}^{\infty} dt\, e^{i\omega t} G^{<}(t) = -iG^{<}(\omega). \tag{2.1.20}$$

This quantity is generally not the dimensionless Fermi function. The occupation is its integral with the inverse Fourier measure,

$$n = \int \frac{d\omega}{2\pi} W_{\mathrm{occ}}(\omega). \tag{2.1.21}$$

In thermal equilibrium, $W_{\mathrm{occ}}(\omega) = f(\omega)A(\omega)$, where A is the spectral function defined below and $f(\omega) = [e^{\beta(\omega-\mu)}+1]^{-1}$. Thus $W_{\mathrm{occ}}/(2\pi)$ is occupied spectral density. For an isolated free mode, $W_{\mathrm{occ}} = 2\pi f(\epsilon_k)\delta(\omega-\epsilon_k)$, which makes the distinction from f explicit. Dividing occupied weight by spectral weight recovers f only on spectral support and in the stated equilibrium regime.

2.1.4 Time ordering and the fermion exchange sign

The time-ordered Green's function is

$$G^{T}(t,t') = -i\langle \mathsf{T} c(t)c^{\dagger}(t')\rangle. \tag{2.1.22}$$

For fermions,

$$\mathsf{T} c(t)c^{\dagger}(t') = \Theta(t-t')c(t)c^{\dagger}(t') - \Theta(t'-t)c^{\dagger}(t')c(t). \tag{2.1.23}$$

For bosons, the exchange has the opposite sign:

$$\mathsf{T} b(t)b^{\dagger}(t') = \Theta(t-t')b(t)b^{\dagger}(t') + \Theta(t'-t)b^{\dagger}(t')b(t). \tag{2.1.24}$$

The symbol Θ is the Heaviside step function. Time ordering places the later time to the left. For fermionic operators, each interchange adds a minus sign; for bosonic operators, it does not. This prescription is an ordering rule for products, and does not say that operators at unequal times can simply be commuted or anticommuted without an additional term.

Applying the ordering rule to the defining expectation value gives the explicit expansion:

$$G^{T}(t,t') = -i\Theta(t-t')\langle c(t)c^{\dagger}(t')\rangle + i\Theta(t'-t)\langle c^{\dagger}(t')c(t)\rangle \tag{2.1.25}$$

$$= \Theta(t-t')G^{>}(t,t') + \Theta(t'-t)G^{<}(t,t'). \tag{2.1.26}$$

Time ordering selects the appropriate operator-ordering branch. In the ket action, the operator that acts first is the rightmost operator; after time ordering it has the smaller time argument. This chronological bookkeeping must be distinguished from the causal support condition of a retarded function. G^{T} generally has nonzero values on both sides of $t=t'$, whereas G^{R} vanishes when its observation time precedes its source time. Equal-time products require a stated one-sided limit; the definition of $G^{<}(t,t)$ in (2.1.18) has no step-function ambiguity.

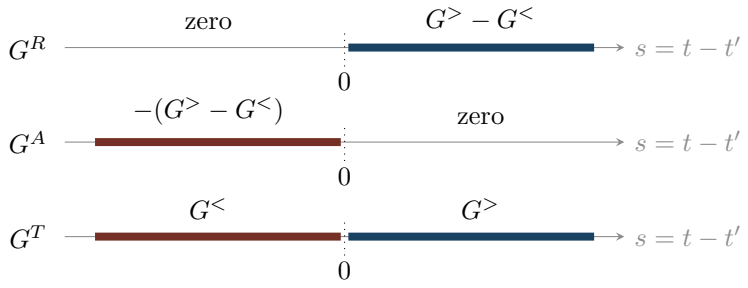


Figure 2.5: Support on the relative-time axis $s = t - t'$. Retarded and advanced functions occupy opposite half-axes, while a time-ordered function uses the greater or lesser component according to the sign of s. These bars indicate support and component selection, not numerical function values.

2.1.5 Retarded and advanced Green's functions

For fermions, define the retarded function by

$$G^R(t,t') = -i\Theta(t-t')\langle\{c(t), c^\dagger(t')\}\rangle \tag{2.1.27}$$

$$= -i\Theta(t-t')\left[\langle c(t)c^\dagger(t')\rangle + \langle c^\dagger(t')c(t)\rangle\right] \tag{2.1.28}$$

$$= \Theta(t-t')\left[G^>(t,t') - G^<(t,t')\right]. \tag{2.1.29}$$

The braces denote an *anticommutator*, $\{X, Y\} = XY + YX$. The support condition is

$$t < t' \quad \Longrightarrow \quad G^R(t,t') = 0. \tag{2.1.30}$$

The observation at t cannot precede the insertion at t' in a retarded propagator. The anticommutator combines particle addition and removal, and its equal-time value supplies the canonical source term in the equation of motion. This single-fermion object should not be confused with a density response: linear response of a density to a scalar potential involves a retarded *commutator of densities*, even though the microscopic particles are fermions.

The advanced function is defined with the opposite time support:

$$G^A(t,t') = +i\Theta(t'-t)\langle\{c(t), c^\dagger(t')\}\rangle \tag{2.1.31}$$

$$= -\Theta(t'-t)\left[G^>(t,t') - G^<(t,t')\right]. \tag{2.1.32}$$

Consequently $G^A(t,t') = 0$ for $t > t'$. The minus sign in the second line is fixed by $G^> - G^< = -i\langle\{c, c^\dagger\}\rangle$; it is not an independent convention. For matrix-valued functions,

$$[G^R_{ji}(t',t)]^* = G^A_{ij}(t,t'), \qquad G^A(\omega) = [G^R(\omega)]^\dagger \tag{2.1.33}$$

in a stationary problem. A conjugation reverses indices and time arguments.

For comparison, a consistent bosonic convention is

$$D^>(t,t') = -i\langle b(t)b^\dagger(t')\rangle, \qquad D^<(t,t') = -i\langle b^\dagger(t')b(t)\rangle, \tag{2.1.34}$$

$$D^R(t,t') = -i\Theta(t-t')\langle[b(t), b^\dagger(t')]\rangle, \qquad D^K(t,t') = -i\langle\{b(t), b^\dagger(t')\}\rangle. \tag{2.1.35}$$

The bosonic lesser sign is negative. With that sign, the component identities retain the same form as the fermionic identities below, while the brackets interchange their roles. Mixing the bosonic lesser sign with a fermionic anticommutator rule would give inconsistent relations.

2.1.6 Matsubara Green's function

Imaginary time occupies a fundamental interval $0 \le \tau < \beta$ of length $\beta = 1/T$, with endpoint values understood as one-sided limits. Imaginary-time evolution is generated by $\mathcal{K} = H - \mu N$:

$$c(\tau) = e^{\tau(H-\mu N)} c\, e^{-\tau(H-\mu N)} = e^{\tau\mathcal{K}} c\, e^{-\tau\mathcal{K}}. \tag{2.1.36}$$

The fermionic Matsubara function is

$$G^M(\tau) = -\langle \mathsf{T}_\tau c(\tau) c^\dagger(0)\rangle. \tag{2.1.37}$$

Here T_τ orders operators in imaginary time. There is no real-time factor of i in this definition. Imaginary-time evolution is not unitary: $e^{\tau\mathcal{K}}$ is the natural continuation associated with the Gibbs weight. The explicit ordering gives

$$G^M(\tau) = \begin{cases} -\langle c(\tau)c^\dagger(0)\rangle, & 0 < \tau < \beta, \\ +\langle c^\dagger(0)c(\tau)\rangle, & -\beta < \tau < 0. \end{cases} \tag{2.1.38}$$

At the origin the canonical algebra fixes the jump:

$$G^M(0^+) - G^M(0^-) = -\langle cc^\dagger + c^\dagger c\rangle = -1. \tag{2.1.39}$$

The thermal boundary condition follows directly from the trace. For $-\beta < \tau < 0$,

$$\rho_{\mathrm{eq}} c(\tau+\beta) = c(\tau)\rho_{\mathrm{eq}}, \tag{2.1.40}$$

$$\langle c(\tau+\beta)c^\dagger\rangle = \mathrm{Tr}[c(\tau)\rho_{\mathrm{eq}}c^\dagger] = \langle c^\dagger c(\tau)\rangle, \tag{2.1.41}$$

$$G^M(\tau+\beta) = -G^M(\tau). \tag{2.1.42}$$

Thus the fermionic extension is antiperiodic. Its Fourier series is

$$G^M(\tau) = \frac{1}{\beta}\sum_{n\in\mathbb{Z}} e^{-i\omega_n\tau} G^M(i\omega_n), \qquad \omega_n = \frac{(2n+1)\pi}{\beta}. \tag{2.1.43}$$

For bosons the corresponding extension is periodic and $\omega_n = 2n\pi/\beta$. Analytic continuation to a real-time retarded function must keep the same energy reference. In particular, a denominator written in terms of $\epsilon_k - \mu$ belongs to the $\mathcal{K}$ reference, whereas our physical real-time free pole is at ϵ_k.

2.1.7 Relations between components

To state the identities, also define the anti-time-ordered component by

$$G^{\bar{T}}(t,t') = \Theta(t'-t)G^>(t,t') + \Theta(t-t')G^<(t,t'). \tag{2.1.44}$$

Anti-time ordering places the earlier real time to the left, including the fermion exchange sign. Away from equal times, $\Theta(t-t') + \Theta(t'-t) = 1$. A symmetric choice $\Theta(0) = 1/2$ may be used for these algebraic identities; occupations and discontinuities are more precisely evaluated by one-sided limits.

The four component identities are

$$G^R = G^T - G^< = G^> - G^{\bar{T}}, \tag{2.1.45}$$

$$G^A = G^T - G^> = G^< - G^{\bar{T}}, \tag{2.1.46}$$

$$G^R - G^A = G^> - G^<, \tag{2.1.47}$$

$$G^> + G^< = G^T + G^{\bar{T}}. \tag{2.1.48}$$

For example, subtracting the lesser function explicitly gives

$$G^T - G^< = \Theta(t-t')G^> + [\Theta(t'-t) - 1]G^< \tag{2.1.49}$$

$$= \Theta(t-t')(G^> - G^<) = G^R. \tag{2.1.50}$$

Similarly,

$$G^T - G^> = [\Theta(t-t') - 1]G^> + \Theta(t'-t)G^< \tag{2.1.51}$$

$$= -\Theta(t'-t)(G^> - G^<) = G^A. \tag{2.1.52}$$

Subtracting these two identities proves the third relation; adding the time-ordered and anti-time-ordered definitions proves the fourth.

It is convenient to name the sum the Keldysh component:

$$G^K \equiv G^> + G^< = -i\langle c(t)c^\dagger(t') - c^\dagger(t')c(t)\rangle = -i\langle [c(t), c^\dagger(t')]\rangle. \tag{2.1.53}$$

For these fermionic definitions G^R contains an anticommutator and G^K contains a commutator. The difference $G^R - G^A$ carries spectral information; the sum G^K also depends on occupation. At equilibrium the two are related by the Fermi function, but they are independent pieces of information for a general nonequilibrium state.

2.1.8 Green's functions of the simplest noninteracting system

Take a set of independent fermionic modes,

$$H_0 = \sum_k \epsilon_k c_k^\dagger c_k. \tag{2.1.54}$$

The label k distinguishes canonical one-particle eigenmodes; it may be a momentum when the system is translationally invariant. Energies ϵ_k are physical energies for evolution with H_0. The chemical potential enters the state through $\rho_{\rm eq} \propto e^{-\beta\sum_k(\epsilon_k-\mu)c_k^\dagger c_k}$. For one mode, with occupations $n_k = 0, 1$, the thermal trace gives

$$\langle c_k^\dagger c_k\rangle = \frac{0 + e^{-\beta(\epsilon_k-\mu)}}{1 + e^{-\beta(\epsilon_k-\mu)}} = \frac{1}{e^{\beta(\epsilon_k-\mu)}+1} \equiv f(\epsilon_k). \tag{2.1.55}$$

The complementary average is $\langle c_k c_k^\dagger\rangle = 1 - f(\epsilon_k)$ by the canonical anticommutator.

The Heisenberg operator is

$$c_k(t) = e^{iH_0 t} c_k e^{-iH_0 t}. \tag{2.1.56}$$

To evaluate it, first compute

$$[H_0, c_k] = \sum_{k'} \epsilon_{k'}[c_{k'}^\dagger c_{k'}, c_k] \tag{2.1.57}$$

$$= -\sum_{k'} \epsilon_{k'} c_{k'}\delta_{kk'} = -\epsilon_k c_k. \tag{2.1.58}$$

The intermediate commutator follows without treating fermions as commuting variables:

$$[c_{k'}^\dagger c_{k'}, c_k] = c_{k'}^\dagger c_{k'} c_k - c_k c_{k'}^\dagger c_{k'} \tag{2.1.59}$$

$$= -c_{k'}^\dagger c_k c_{k'} - (\delta_{kk'} - c_{k'}^\dagger c_k)c_{k'} \tag{2.1.60}$$

$$= -\delta_{kk'} c_{k'}. \tag{2.1.61}$$

The Heisenberg equation $i\partial_t c_k(t) = [c_k(t), H_0]$ is therefore $\partial_t c_k(t) = -i\epsilon_k c_k(t)$, with initial condition $c_k(0) = c_k$. Its solution is

$$c_k(t) = c_k e^{-i\epsilon_k t}. \tag{2.1.62}$$

Hermitian conjugation gives

$$c_k^\dagger(t) = c_k^\dagger e^{+i\epsilon_k t}. \tag{2.1.63}$$

We denote these *free* Green's functions by the lower-case letter g.

The greater function follows in two steps:

$$g_k^>(t) = -i\langle c_k(t)c_k^\dagger(0)\rangle = -ie^{-i\epsilon_k t}\langle c_k c_k^\dagger\rangle \tag{2.1.64}$$
$$= -i[1-f(\epsilon_k)]e^{-i\epsilon_k t}. \tag{2.1.65}$$

The lesser function similarly gives

$$g_k^<(t) = i\langle c_k^\dagger(0)c_k(t)\rangle = ie^{-i\epsilon_k t}\langle c_k^\dagger c_k\rangle \tag{2.1.66}$$
$$= ie^{-i\epsilon_k t}f(\epsilon_k). \tag{2.1.67}$$

The exponential specifies coherent mode evolution; the prefactors specify whether the state is empty enough to accept a particle or occupied enough to remove one. At $t=0$, $-ig_k^<(0) = f(\epsilon_k)$ and $i[g_k^>(0) - g_k^<(0)] = 1$, providing direct sign and normalization checks.

The time-ordered function retains both time sectors:

$$g_k^T(t) = -i\langle \mathsf{T}c_k(t)c_k^\dagger(0)\rangle \tag{2.1.68}$$
$$= \Theta(t)g_k^>(t) + \Theta(-t)g_k^<(t) \tag{2.1.69}$$
$$= \{-i\Theta(t)[1-f(\epsilon_k)] + i\Theta(-t)f(\epsilon_k)\}\, e^{-i\epsilon_k t}. \tag{2.1.70}$$

The retarded function is

$$g_k^R(t) = \Theta(t)[g_k^>(t) - g_k^<(t)] \tag{2.1.71}$$
$$= -i\Theta(t)e^{-i\epsilon_k t}. \tag{2.1.72}$$

It does not depend on $f(\epsilon_k)$. It is the propagator of a one-particle excitation. The cancellation uses $[1-f]+f=1$. For this quadratic diagonal problem the anticommutator is a state-independent number times the free phase. Interacting spectral functions generally depend on the state through their excitation energies and matrix elements; the occupation independence found here should not be generalized to arbitrary interacting systems.

2.1.9 The retarded pole and spectral normalization

Define the spectral function in the convention of this lecture by

$$A_k(\omega) = -2\,\mathrm{Im}\, g_k^R(\omega) = i[g_k^R(\omega) - g_k^A(\omega)]. \tag{2.1.73}$$

The Fourier calculation is

$$g_k^R(\omega) = \int_{-\infty}^{\infty} dt\, e^{i\omega t} g_k^R(t) \tag{2.1.74}$$
$$= \lim_{\eta_0\downarrow 0}\int_0^\infty dt\,(-i)e^{i(\omega-\epsilon_k+i\eta_0)t} \tag{2.1.75}$$
$$= \frac{1}{\omega-\epsilon_k+i0^+}. \tag{2.1.76}$$

The regulator $\eta_0 > 0$ multiplies the integrand by $e^{-\eta_0 t}$, so the upper endpoint vanishes. At finite regulator, the antiderivative gives

$$-i\left[\frac{e^{i(\omega-\epsilon_k+i\eta_0)t}}{i(\omega-\epsilon_k+i\eta_0)}\right]_0^\infty = \frac{1}{\omega-\epsilon_k+i\eta_0}. \tag{2.1.77}$$

The limit is distributional:

$$\frac{1}{x+i0^+} = \mathcal{P}\frac{1}{x} - i\pi\delta(x), \qquad \frac{1}{x-i0^+} = \mathcal{P}\frac{1}{x} + i\pi\delta(x). \tag{2.1.78}$$

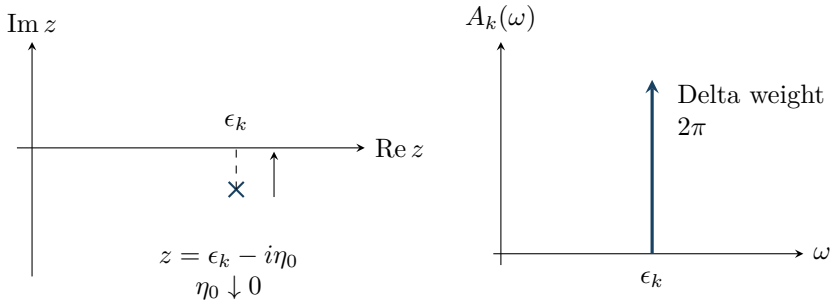


Figure 2.6: A free retarded pole and its spectral weight. The pole approaches the real axis from below as $\eta_0 \downarrow 0$. The spectral delta is drawn as an arrow, whose label gives its integrated weight 2π; the arrow height is not the value of a delta distribution.

The lower-half-plane retarded pole is equivalent to positive-time support. The infinitesimal enforces a boundary condition; it is not a physical lifetime. Taking the imaginary part of the pole yields the corresponding spectral delta:

$$A_k(\omega) = 2\pi\delta(\omega - \epsilon_k). \tag{2.1.79}$$

The density of states of this one canonical mode is $\nu_k(\omega) = A_k(\omega)/(2\pi) = \delta(\omega - \epsilon_k)$, and

$$\int \frac{d\omega}{2\pi} A_k(\omega) = 1. \tag{2.1.80}$$

For a collection of modes one sums these delta functions. For a local continuum field, the normalization instead includes its density of modes per length; the unit sum rule above applies to a canonical mode.

For later transport calculations, Fourier transforming the lesser and greater functions separately gives

$$g_k^<(\omega) = if(\epsilon_k) \int dt\, e^{i(\omega-\epsilon_k)t} = 2\pi i f(\epsilon_k)\delta(\omega - \epsilon_k), \tag{2.1.81}$$

$$g_k^>(\omega) = -i[1 - f(\epsilon_k)] \int dt\, e^{i(\omega-\epsilon_k)t} = -2\pi i[1 - f(\epsilon_k)]\delta(\omega - \epsilon_k). \tag{2.1.82}$$

The advanced and time-ordered frequency functions are

$$g_k^A(\omega) = \frac{1}{\omega - \epsilon_k - i0^+}, \tag{2.1.83}$$

$$g_k^T(\omega) = \frac{1 - f(\epsilon_k)}{\omega - \epsilon_k + i0^+} + \frac{f(\epsilon_k)}{\omega - \epsilon_k - i0^+}. \tag{2.1.84}$$

Thus the time-ordered function contains both pole prescriptions. The complete free equilibrium set obeys

$$g_k^< = if A_k, \qquad g_k^> = -i(1 - f)A_k, \qquad g_k^K = (1 - 2f)(g_k^R - g_k^A). \tag{2.1.85}$$

On the delta support, $f(\omega) = f(\epsilon_k)$, which explains why either argument may be used here. These occupation relations require the same thermal state and energy reference; a biased multi-reservoir system does not generally have one global scalar Fermi function.

2.1.10 Review of the equation-of-motion method

A discrete level coupled to a free lead illustrates how a sharp excitation acquires a width. This is a quadratic resonant-level model, with no dot-dot interaction. The dot annihilation operator is d, the lead modes are c_k, and all are canonical fermions in the same Fock space: $\{d, d^\dagger\} = 1$, $\{c_k, c_{k'}^\dagger\} = \delta_{kk'}$, and $\{c_k, d^\dagger\} = 0$. The three Hamiltonian contributions are

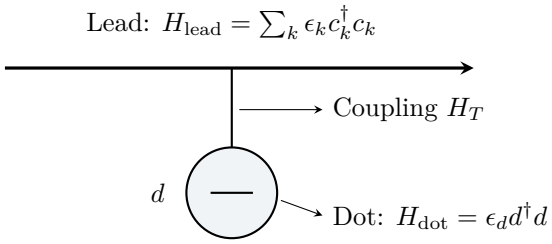


Figure 2.7: A free lead coupled locally to a discrete dot level. The coupling connects lead modes c_k to the dot operator d; the horizontal arrow belongs to the lead sketch and does not add a second reservoir.

$$H_{\text{lead}} = \sum_k \epsilon_k c_k^\dagger c_k, \tag{2.1.86}$$

$$H_T = \sum_k \left(V_k c_k^\dagger d + V_k^* d^\dagger c_k \right), \tag{2.1.87}$$

$$H_{\text{dot}} = \epsilon_d d^\dagger d. \tag{2.1.88}$$

The total Hamiltonian is $H = H_{\text{lead}} + H_T + H_{\text{dot}}$. The amplitudes V_k may be complex; choosing them real gives a useful phase-choice specialization. Lead-dot hopping has energy units because c_k and d are normalized discrete operators. This differs from the continuum contact hopping introduced later.

How can we solve for the retarded Green's function? For the dot, define

$$G_d^R(t) = -i\Theta(t)\langle\{d(t), d^\dagger(0)\}\rangle. \tag{2.1.89}$$

This function reveals the impurity spectral function. The Heisenberg equation of motion is

$$i\frac{dA(t)}{dt} = [A(t), H]. \tag{2.1.90}$$

The operator $A(t)$ is in the Heisenberg picture. Equation (2.1.90) assumes no explicit time dependence of the Schrödinger operator. If such a dependence is present, add $i(\partial_t A)_{\text{explicit}}$. Here the Hamiltonian is time independent, and the stationary retarded function can be Fourier transformed.

Use the inverse transform

$$G_d^R(t) = \int_{-\infty}^{\infty} \frac{d\omega'}{2\pi} e^{-i\omega' t} G_d^R(\omega'). \tag{2.1.91}$$

Differentiate both sides with respect to t.

$$\begin{aligned}\int \frac{d\omega'}{2\pi}(-i\omega')e^{-i\omega' t} G_d^R(\omega') = &-i\delta(t)\langle\{d(t), d^\dagger(0)\}\rangle \\ &- i\Theta(t)\left\langle\left\{-i[d(t), H], d^\dagger(0)\right\}\right\rangle.\end{aligned} \tag{2.1.92}$$

The first term is from $d\Theta(t)/dt = \delta(t)$; the second differentiates the annihilation operator. The delta function evaluates the anticommutator at equal times, so

$$\delta(t)\langle\{d(t), d^\dagger(0)\}\rangle = \delta(t). \tag{2.1.93}$$

This is the origin of the inhomogeneous unit source in the frequency equation. Dropping the derivative of the step function would incorrectly replace the resolvent equation by a homogeneous one.

The commutator entering the second term is

$$[d, H] = \epsilon_d d + \sum_k V_k^* c_k. \tag{2.1.94}$$

For fermionic a, b, c, the useful algebraic identity is $[a, bc] = \{a, b\}c - b\{a, c\}$. Applying it term by term,

$$[d, d^\dagger d] = d, \qquad [d, c_k^\dagger c_k] = 0, \tag{2.1.95}$$

$$[d, c_k^\dagger d] = 0, \qquad [d, d^\dagger c_k] = c_k. \tag{2.1.96}$$

This fixes both the coefficient and the complex conjugation in (2.1.94). Multiply by $e^{i\omega t}$ and integrate over t on both sides. The transform is evaluated using

$$\int_{-\infty}^{\infty} dt\, e^{i(\omega-\omega')t} = 2\pi\delta(\omega - \omega'), \qquad \int dt\, e^{i\omega t}\delta(t) = 1. \tag{2.1.97}$$

Equivalently, multiply (2.1.92) by i first. The Fourier transform of $i\partial_t G_d^R$ is ωG_d^R, with the retarded boundary condition understood. The first frequency equation is

$$\omega G_d^R(\omega) = 1 + \epsilon_d G_d^R(\omega) + \sum_k V_k^* G_{kd}^R(\omega). \tag{2.1.98}$$

The mixed function is defined by

$$G_{kd}^R(t) = -i\Theta(t)\langle\{c_k(t), d^\dagger(0)\}\rangle. \tag{2.1.99}$$

Its first index refers to lead annihilation and its second to dot creation. The same calculation now has no equal-time source because $\{c_k, d^\dagger\} = 0$. The relevant operator commutator is

$$[c_k, H] = \epsilon_k c_k + V_k d. \tag{2.1.100}$$

Thus G_{kd}^R describes an amplitude that leaves the initial dot insertion and is observed in a lead mode, retaining the operator-index order. The second frequency equation is

$$\omega G_{kd}^R(\omega) = \epsilon_k G_{kd}^R(\omega) + V_k G_d^R(\omega). \tag{2.1.101}$$

Both calculations are examples of the general fermionic retarded EOM. For

$$G_{A,B}^R(t) = -i\Theta(t)\langle\{A(t), B(0)\}\rangle, \tag{2.1.102}$$

where B need not itself be an annihilation operator,

$$i\partial_t G_{A,B}^R(t) = \delta(t)\langle\{A, B\}\rangle + G_{[A,H],B}^R(t), \tag{2.1.103}$$

$$\omega G_{A,B}^R(\omega) = \langle\{A, B\}\rangle + G_{[A,H],B}^R(\omega). \tag{2.1.104}$$

For a quadratic Hamiltonian the commutator stays linear in fermion operators, so the equations close. An interaction generally produces higher-order operators and a hierarchy; closing that hierarchy then requires an explicit approximation or additional information.

2.1.11 Eliminating the lead and identifying the self-energy

Solve the two coupled equations by first rearranging them:

$$(\omega - \epsilon_d + i0^+)G_d^R(\omega) = 1 + \sum_k V_k^* G_{kd}^R(\omega), \tag{2.1.105}$$

$$(\omega - \epsilon_k + i0^+)G_{kd}^R(\omega) = V_k G_d^R(\omega). \tag{2.1.106}$$

Substitute the second equation into the first to eliminate each lead mode. Keep the retarded prescription during elimination. The intermediate steps are

$$G^R_{kd}(\omega) = \frac{V_k}{\omega - \epsilon_k + i0^+} G^R_d(\omega), \tag{2.1.107}$$

$$(\omega - \epsilon_d + i0^+) G^R_d(\omega) = 1 + \sum_k \frac{|V_k|^2}{\omega - \epsilon_k + i0^+} G^R_d(\omega), \tag{2.1.108}$$

$$\left[\omega - \epsilon_d + i0^+ - \sum_k \frac{|V_k|^2}{\omega - \epsilon_k + i0^+}\right] G^R_d(\omega) = 1. \tag{2.1.109}$$

Therefore the dot resolvent is

$$G^R_d(\omega) = \frac{1}{\omega - \epsilon_d + i0^+ - \Sigma^R_d(\omega)}. \tag{2.1.110}$$

The lead-induced self-energy is

$$\Sigma^R_d(\omega) = \sum_k \frac{|V_k|^2}{\omega - \epsilon_k + i0^+}. \tag{2.1.111}$$

The word self-energy here means the effect of eliminating lead degrees of freedom. It does not imply electron-electron interactions. The denominator describes leaving the dot, propagating in a lead mode, and returning to the dot. Complex hopping amplitudes appear as $V_k^* V_k = |V_k|^2$, so an arbitrary phase convention for each lead mode cannot change the dot spectrum.

Using (2.1.78), split the self-energy into a real shift and a nonnegative width:

$$\Sigma^R_d(\omega) = \Lambda_d(\omega) - i\Gamma_d(\omega), \tag{2.1.112}$$

$$\Lambda_d(\omega) = \mathcal{P} \sum_k \frac{|V_k|^2}{\omega - \epsilon_k}, \tag{2.1.113}$$

$$\Gamma_d(\omega) = \pi \sum_k |V_k|^2 \delta(\omega - \epsilon_k) \geq 0. \tag{2.1.114}$$

For a continuum of lead modes with density $\nu_{\text{lead}}(E) = \sum_k \delta(E - \epsilon_k)$ and a coupling depending smoothly on energy,

$$\Gamma_d(\omega) = \pi |V(\omega)|^2 \nu_{\text{lead}}(\omega). \tag{2.1.115}$$

The real part is a principal-value integral over the actual band. It cannot be discarded simply by calling a lead "flat". A symmetric wide-band model about the chosen energy reference may make it negligible locally, or a nearly constant value may be absorbed into a renormalized level energy.

For example, a flat hybridization with bare constant width Γ_0 on the finite interval $-D < E < D$ has

$$\Lambda_d(\omega) = \frac{\Gamma_0}{\pi} \ln \left|\frac{\omega + D}{\omega - D}\right|, \qquad \Gamma_d(\omega) = \Gamma_0 \, \Theta(D - |\omega|). \tag{2.1.116}$$

For $|\omega| \ll D$, $\Lambda_d \simeq 2\Gamma_0 \omega/(\pi D)$ is small. This example explains the wide-band approximation locally; behavior near a band edge still requires the finite-band expression. A finite isolated set of lead levels instead yields a discrete hybridized spectrum and recurrences, not irreversible exponential decay for all times.

In a constant-width approximation, retain or absorb the real shift by defining $\widetilde{\epsilon}_d = \epsilon_d + \Lambda_d$. Then

$$G^R_d(\omega) \approx \frac{1}{\omega - \widetilde{\epsilon}_d + i\Gamma_d}, \qquad \Gamma_d > 0. \tag{2.1.117}$$

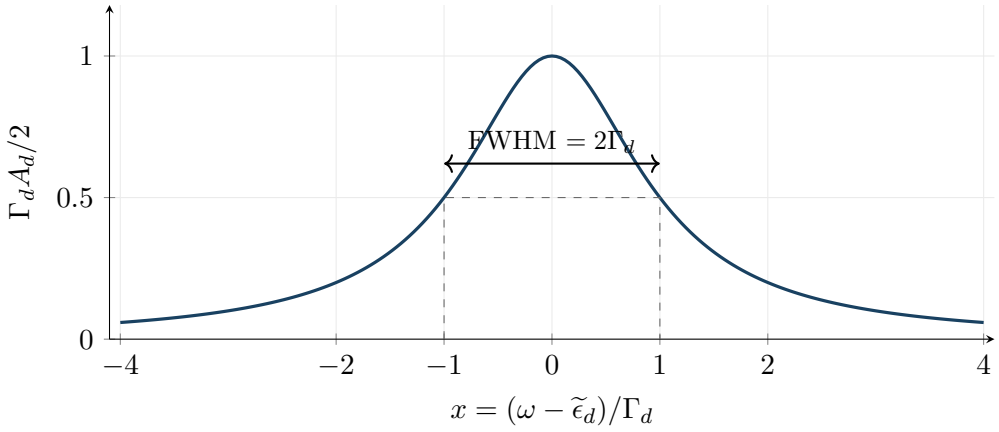


Figure 2.8: Broadening in the constant-width model. The curve is $\Gamma_d A_d/2 = 1/(1+x^2)$ with $x = (\omega - \widetilde{\epsilon}_d)/\Gamma_d$. Half maximum occurs at $x = \pm 1$, so the full width in energy is $2\Gamma_d$. The curve is an analytic model, not experimental data.

This approximation is appropriate when the hybridization and real shift vary slowly across the resonance. The full energy-dependent formula remains (2.1.110). The spectral function is

$$A_d(\omega) = -2\,\mathrm{Im}\,G_d^R(\omega). \tag{2.1.118}$$

In the constant-width regime this gives

$$A_d(\omega) \approx \frac{2\Gamma_d}{(\omega - \widetilde{\epsilon}_d)^2 + \Gamma_d^2}. \tag{2.1.119}$$

Compare this resonance to the isolated-dot spectral weight $2\pi\delta(\omega - \epsilon_d)$. The coupling produces *level broadening*.

Three checks fix the normalization and physical interpretation. First,

$$\int_{-\infty}^{\infty} \frac{d\omega}{2\pi} A_d(\omega) = \int_{-\infty}^{\infty} \frac{dx}{\pi(1+x^2)} = 1. \tag{2.1.120}$$

Second,

$$\lim_{\Gamma_d \downarrow 0} \frac{2\Gamma_d}{(\omega - \widetilde{\epsilon}_d)^2 + \Gamma_d^2} = 2\pi\delta(\omega - \widetilde{\epsilon}_d) \tag{2.1.121}$$

as a distribution. Third, inverse transformation of the single pole yields

$$G_d^R(t) = -i\Theta(t) e^{-i\widetilde{\epsilon}_d t} e^{-\Gamma_d t}. \tag{2.1.122}$$

Thus Γ_d is the spectral half-width and the inverse amplitude-decay time. The full width is $2\Gamma_d$; the corresponding probability in this single-pole decay model falls as $e^{-2\Gamma_d t}$. With units restored, the amplitude and probability decay times are $\hbar/\Gamma_d$ and $\hbar/(2\Gamma_d)$, respectively. This probability statement concerns the decaying single-particle amplitude; an equilibrium dot occupation also includes particles supplied by the lead. The dot width is an energy and must not be confused with the dimensionless contact strength used in the later single-barrier problem.

2.2 Keldysh formalism: retaining both directions of evolution

2.2.1 The equilibrium phase simplification and its limits

We first examine the evolution operators that enter an equilibrium expectation value or correlation function.

$$\langle O_H(t)\rangle = \mathrm{Tr}[\rho_0 U(t_0,t) O U(t,t_0)]\,. \tag{2.2.1}$$

$$U(t,t_0) = e^{-iH(t-t_0)} \qquad (H \text{ independent of time}). \tag{2.2.2}$$

$U(t,t_0)$ propagates the ket forward from the initial time t_0 to t. The remote-past notation $t_0 \to -\infty$ denotes a preparation before the measurement; an adiabatic switch-on is a further assumption.

$$U(t_0,t) = U(t,t_0)^\dagger, \qquad U(t_0,t)U(t,t_0) = 1. \tag{2.2.3}$$

For a time-dependent Hamiltonian, forward evolution is

$$U(t,t_0) = \mathsf{T}\exp\left[-i\int_{t_0}^{t} ds\, H(s)\right], \qquad t \geq t_0. \tag{2.2.4}$$

Backward evolution is defined by the adjoint,

$$U(t_0,t) = U(t,t_0)^\dagger = \overline{\mathsf{T}}\exp\left[+i\int_{t_0}^{t} ds\, H(s)\right]. \tag{2.2.5}$$

Both directions obey $U(t_2,t_1)U(t_1,t_0) = U(t_2,t_0)$. The bra and the ket evolve in opposite directions. This is exact unitary quantum mechanics and remains valid away from equilibrium.

A useful equilibrium simplification is available when the prepared *pure state* returns to itself up to a phase under the complete adiabatic evolution. The stationarity condition $[\rho_0, H] = 0$ by itself does not establish this single-phase statement for a thermal mixture.

$$\begin{gathered} U(t_f,t_0)|\Psi_0\rangle = e^{i\phi}|\Psi_0\rangle, \qquad \rho_0 = |\Psi_0\rangle\langle\Psi_0|, \\ (t_0,t_f) \to (-\infty,+\infty). \end{gathered} \tag{2.2.6}$$

Here ϕ is real and depends on the switching history. For an eigenvector of the unitary $U(t_f,t_0)$, the corresponding bra obeys $\langle\Psi_0|U(t_f,t_0) = e^{i\phi}\langle\Psi_0|$. Hence the composition rule gives $\langle\Psi_0|U(t_f,t) = e^{i\phi}\langle\Psi_0|U(t_0,t)$. By contrast, a diagonal thermal state $\rho_{\mathrm{eq}} = \sum_n p_n|n\rangle\langle n|$ can be stationary while $U|n\rangle = e^{-iE_n(t_f-t_0)}|n\rangle$ has a different phase for every occupied energy.

$$\langle\Psi_0|U(t_0,t) = \frac{\langle\Psi_0|U(t_f,t)}{\langle\Psi_0|U(t_f,t_0)|\Psi_0\rangle}. \tag{2.2.7}$$

$$\langle\Psi_0|U(t_0,t) = e^{-i\phi}\langle\Psi_0|U(t_f,t). \tag{2.2.8}$$

$$\langle O_H(t)\rangle = e^{-i\phi}\langle\Psi_0|U(t_f,t)OU(t,t_0)|\Psi_0\rangle. \tag{2.2.9}$$

Read the product from right to left: evolve from t_0 to t, insert O, and continue from t to t_f. Under the stated phase condition, the entire expectation can therefore be drawn on a single forward time axis.

For two insertions with $t_1 < t_2$, the corresponding time-ordered numerator is

$$\langle\Psi_0|U(t_f,t_2)O_2U(t_2,t_1)O_1U(t_1,t_0)|\Psi_0\rangle. \tag{2.2.10}$$

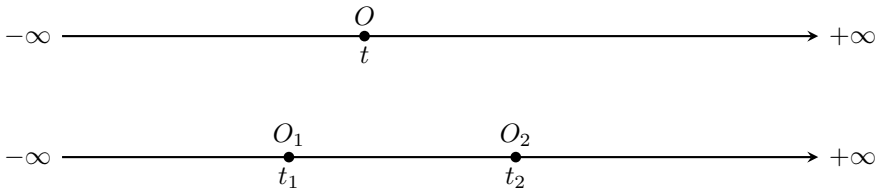


Figure 2.9: Single-axis equilibrium evolution under the pure-state phase condition. The upper line contains one insertion; the lower line retains the two insertions in their real-time order.

2.2.2 Why a closed time contour is needed

$$\rho_{\mathrm{eq}} = \frac{e^{-\beta(H-\mu N)}}{Z}, \qquad Z = \mathrm{Tr}\, e^{-\beta(H-\mu N)}. \tag{2.2.11}$$

A driven or biased preparation is generally not a global Gibbs state of the coupled Hamiltonian with a single chemical potential.

The pure-state phase condition can fail:

$$U(t_f, t_0)|\Psi_0\rangle \not\propto |\Psi_0\rangle. \tag{2.2.12}$$

Consequently the preceding reduction to one forward axis is unavailable, although the two evolution operators remain well defined.

$$\langle O_H(t)\rangle = \mathrm{Tr}[\rho_0 U(t_0, t) O U(t, t_0)]\,. \tag{2.2.13}$$

ρ_0 specifies the initial state. The factor on its right that acts first on a ket, $U(t, t_0)$, supplies the forward branch; $U(t_0, t)$ supplies the backward branch.

2.2.2.1 Preparation, switch-on, and the two branches

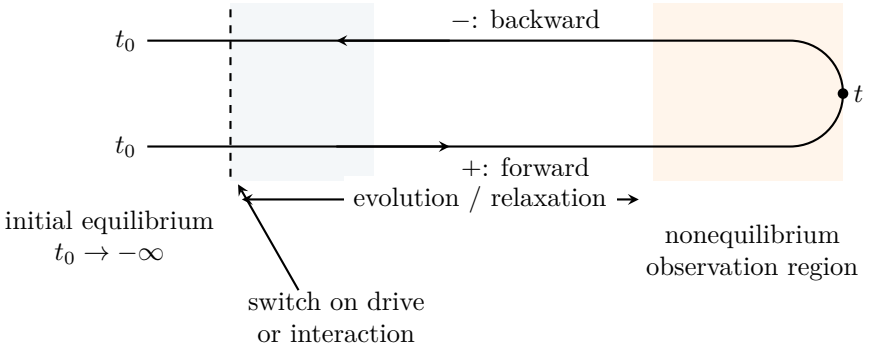


Figure 2.10: A closed contour for a one-time expectation value. Evolution starts in the prepared state, passes the switch-on time, reaches the observation at the turn, and returns on the backward branch. A relaxation interval precedes the nonequilibrium observation region.

Prepare an equilibrium reference at t_0 with an initial Hamiltonian H_{eq}. In the contact problem this means that each disconnected reservoir is separately equilibrated.

At a chosen switch-on time, turn on the drive or the coupling, after which the Hamiltonian may be denoted $H_{\mathrm{neq}}(t)$.

The switch-on transient precedes the later nonequilibrium region of interest. Taking a long waiting time is useful when a steady state exists; its existence and the disappearance of initial memory require properties of the actual reservoirs and spectrum.

The two real branches exactly represent unitary expectations for a specified ρ_0. A Gaussian disconnected reference permits the Wick expansion used later. A general correlated thermal preparation may require an imaginary-time branch or explicit initial-correlation terms in its perturbative representation. Merely writing $t_0 = -\infty$ does not remove such terms or prove decay of a bound-state contribution.

2.2.2.2 Two operator insertions

$$\langle A_H(t)B_H(t')\rangle = \mathrm{Tr}[\rho_0 A_H(t)B_H(t')]. \tag{2.2.14}$$

$$\begin{aligned}\langle A_H(t)B_H(t')\rangle = \mathrm{Tr}\big[&\rho_0 S(t_0,t)A_I(t)S(t,t') \\ &\times B_I(t')S(t',t_0)\big].\end{aligned} \tag{2.2.15}$$

$S(t,t')$ evolves with the interaction-picture perturbation $H_{T,I}$, rather than the reference Hamiltonian H_0.

For $H = H_0 + H_T$, define $A_I(t) = e^{iH_0(t-t_0)}Ae^{-iH_0(t-t_0)}$. For $t \geq t'$,

$$S(t,t') = \mathsf{T}\exp\left[-i\int_{t'}^{t} ds\, H_{T,I}(s)\right]. \tag{2.2.16}$$

For $t < t'$, the backward evolution is

$$S(t,t') = S(t',t)^\dagger = \overline{\mathsf{T}}\exp\left[+i\int_{t}^{t'} ds\, H_{T,I}(s)\right]. \tag{2.2.17}$$

Inserting $A_H(t) = S(t_0,t)A_I(t)S(t,t_0)$ for each operator and combining $S(t,t_0)S(t_0,t') = S(t,t')$ yields (2.2.15). The order of factors fixes the placement of the insertions on a contour.

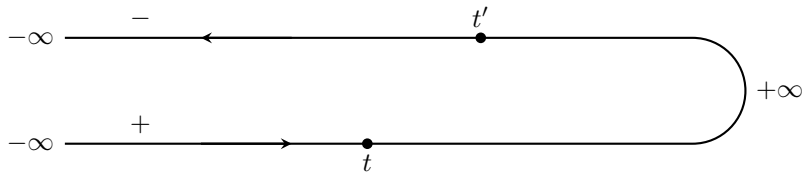


Figure 2.11: Here t lies on the forward branch and t' on the backward branch. With A at t and B at t', contour ordering gives BA (and a minus sign for a fermionic interchange). To represent the ordered product $A(t)B(t')$ in (2.2.15), put A on the backward branch and B on the forward branch.

2.2.3 Contour ordering and the four real-time components

$$G_{ij}^{\kappa\kappa'}(t,t') = -i\left\langle \mathsf{T}_C\psi_i(t^\kappa)\psi_j^\dagger((t')^{\kappa'})\right\rangle. \tag{2.2.18}$$

The branch indices $\kappa, \kappa' \in \{+,-\}$ are contour indices. They are independent of the physical field indices i, j.

$$\mathsf{T}_C\psi_i(t^\kappa)\psi_j^\dagger((t')^{\kappa'}) = \psi_i(t^\kappa)\psi_j^\dagger((t')^{\kappa'}), \qquad t^\kappa >_C (t')^{\kappa'}. \tag{2.2.19}$$

$$\mathsf{T}_C\psi_i(t^\kappa)\psi_j^\dagger((t')^{\kappa'}) = -\psi_j^\dagger((t')^{\kappa'})\psi_i(t^\kappa), \qquad t^\kappa <_C (t')^{\kappa'}. \tag{2.2.20}$$

$>_C$ means downstream, or later *along the contour*; $<_C$ means upstream. On the return branch, later contour position corresponds to smaller real time.

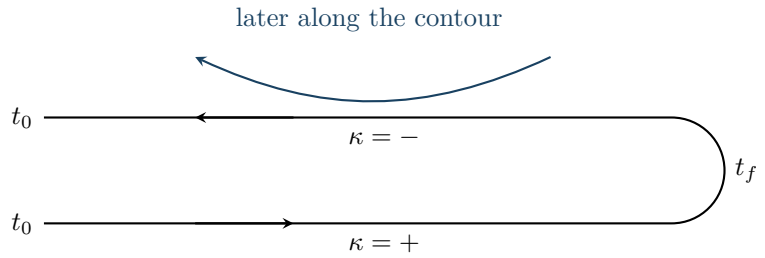


Figure 2.12: Contour order follows the arrows: all points on the return branch are later along the contour than every point on the forward branch. Both left endpoints have real time t_0; the right turn is t_f.

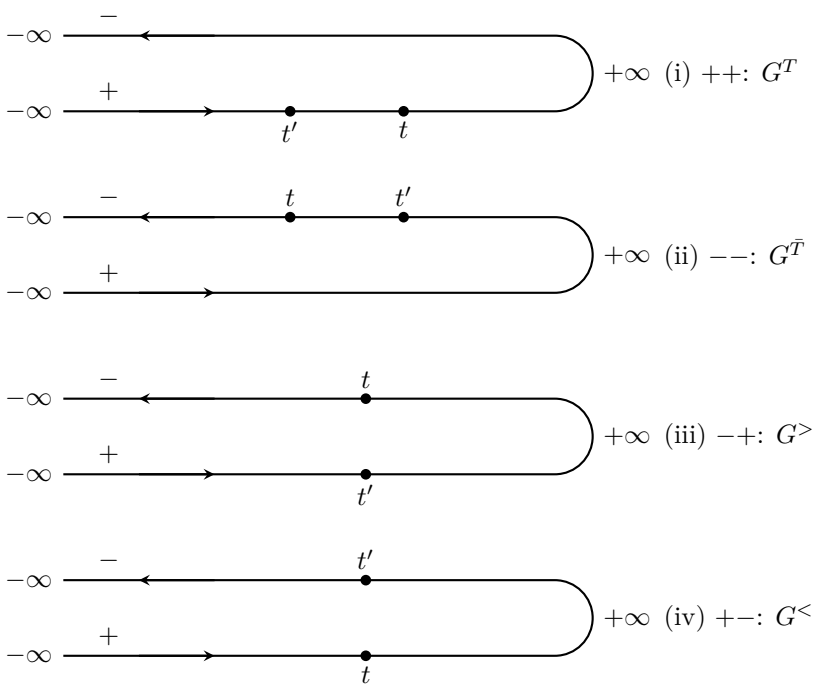


Figure 2.13: The four insertion patterns. Same-branch panels implement time ordering or anti-time ordering; cross-branch panels encode greater and lesser functions regardless of the real-time comparison. Every forward arrow points right and every backward arrow points left.

$$G_{ij}^{++}(t,t') = -i\langle \mathsf{T}\psi_i(t)\psi_j^\dagger(t')\rangle = G_{ij}^{T}(t,t'). \tag{2.2.21}$$

For the $--$ panel, the two insertions lie on the return branch; its reversed direction implements anti-time ordering.

$$G_{ij}^{--}(t,t') = -i\langle \overline{\mathsf{T}}\psi_i(t)\psi_j^\dagger(t')\rangle = G_{ij}^{\bar{T}}(t,t'). \tag{2.2.22}$$

For the $-+$ panel, $\psi_i(t^-)$ is later on the contour than $\psi_j^\dagger((t')^+)$.

$$G_{ij}^{-+}(t,t') = -i\langle \psi_i(t)\psi_j^\dagger(t')\rangle = G_{ij}^{>}(t,t'). \tag{2.2.23}$$

For the $+-$ panel, contour ordering exchanges the two fermions and contributes one minus sign.

$$G_{ij}^{+-}(t,t') = +i\langle \psi_j^\dagger(t')\psi_i(t)\rangle = G_{ij}^{<}(t,t'). \tag{2.2.24}$$

$$\mathbb{G}_C = \begin{pmatrix} G^{++} & G^{+-} \\ G^{-+} & G^{--} \end{pmatrix} = \begin{pmatrix} G^T & G^< \\ G^> & G^{\bar{T}} \end{pmatrix}, \qquad \text{branch order } (+,-). \tag{2.2.25}$$

The four branch *combinations* also appear in perturbative expansions. There are two contour branches and four ways to assign the two external insertions.

$$S_C = \mathsf{T}_C \exp\left[-i \int_C dz\, H_{T,I}(z)\right]. \tag{2.2.26}$$

$$\mathcal{C}_{12} = \left\langle \mathsf{T}_C \psi_{1,I}(z_1) \psi^\dagger_{2,I}(z_2) S_C \right\rangle_0. \tag{2.2.27}$$

Here $\mathcal{C}$ is a raw correlator, so the corresponding Green's function is $-i\mathcal{C}$. For a closed unitary contour with the same Hamiltonian on both branches, $\langle S_C\rangle_0 = 1$. If a generating construction has unequal branch sources, its normalization must be retained. Expanding the displayed exponential gives the zeroth-order term

$$\mathcal{C}^{(0)}_{12} = \langle \mathsf{T}_C \psi_{1,I}(z_1) \psi^\dagger_{2,I}(z_2)\rangle_0. \tag{2.2.28}$$

$$\mathcal{C}^{(1)}_{12} = -i \int_C dz\, \left\langle \mathsf{T}_C \psi_{1,I}(z_1) \psi^\dagger_{2,I}(z_2) H_{T,I}(z) \right\rangle_0. \tag{2.2.29}$$

$$\begin{aligned} \mathcal{C}^{(2)}_{12} = \frac{(-i)^2}{2!} \int_C dz \int_C dz' \,\big\langle \mathsf{T}_C \psi_{1,I}(z_1) \psi^\dagger_{2,I}(z_2) \\ \times H_{T,I}(z) H_{T,I}(z') \big\rangle_0. \end{aligned} \tag{2.2.30}$$

Thus $\mathcal{C} = \mathcal{C}^{(0)} + \mathcal{C}^{(1)} + \mathcal{C}^{(2)} + \cdots$. The powers of $-i$ and the factorial follow directly from the exponential. Each internal contour integration includes both branches:

$$\int_C dz\, F(z) = \sum_{\kappa=\pm 1} \kappa \int_{t_0}^{t_f} dt\, F(t^\kappa). \tag{2.2.31}$$

The minus sign for the return branch is an orientation sign, distinct from a fermionic exchange sign.

2.2.4 Rotation, Dyson equation, and Langreth rules

$$\check{G} = \begin{pmatrix} G^R & G^K \\ 0 & G^A \end{pmatrix}. \tag{2.2.32}$$

$$\check{G} = \mathsf{L}_C \mathsf{m}_C \mathbb{G}_C \mathsf{L}_C^\dagger, \quad \mathsf{L}_C = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}, \quad \mathsf{m}_C = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{2.2.33}$$

The contour metric m_C records the reversed integration direction on the backward branch. It acts in contour space and is unrelated to physical spin or the reservoir labels L, R.

Using $G^T + G^{\bar{T}} = G^> + G^<$, multiply the three matrices explicitly:

$$\mathsf{L}_C \mathsf{m}_C \mathbb{G}_C = \frac{1}{\sqrt{2}} \begin{pmatrix} G^T + G^> & G^< + G^{\bar{T}} \\ G^T - G^> & G^< - G^{\bar{T}} \end{pmatrix}, \tag{2.2.34}$$

$$\check{G} = \frac{1}{2} \begin{pmatrix} G^T + G^> - G^< - G^{\bar{T}} & G^T + G^> + G^< + G^{\bar{T}} \\ G^T - G^> - G^< + G^{\bar{T}} & G^T - G^> + G^< - G^{\bar{T}} \end{pmatrix} \tag{2.2.35}$$

$$= \begin{pmatrix} G^R & G^K \\ 0 & G^A \end{pmatrix}. \tag{2.2.36}$$

The zero entry is the component identity, rather than an approximation.

$$G^K_{ij}(t,t') = G^>_{ij}(t,t') + G^<_{ij}(t,t'). \tag{2.2.37}$$

$$G^K_{ij}(t,t') = -i\langle[\psi_i(t), \psi^\dagger_j(t')]\rangle. \tag{2.2.38}$$

Indeed, $-i\langle\psi_i\psi^\dagger_j\rangle + i\langle\psi^\dagger_j\psi_i\rangle = -i\langle[\psi_i, \psi^\dagger_j]\rangle$. The fermionic retarded function contains an anticommutator; the fermionic Keldysh function contains a commutator and carries occupation information.

$$\check{G} = \check{g} + \check{g} \star \check{\Sigma} \star \check{G}. \tag{2.2.39}$$

g is a known reference Green's function. The kernel Σ may be a deterministic one-body hopping relative to a disconnected reference; its appearance in Dyson's equation does not by itself make it an interaction self-energy.

The symbol $\star$ includes real-time integration and ordered multiplication in any internal physical indices:

$$(A \star B)_{ij}(t,t') = \sum_m \int_{t_0}^{t_f} ds\, A_{im}(t,s) B_{mj}(s,t'). \tag{2.2.40}$$

Before rotation, contour convolution obeys $\mathbb{C}_C = \mathbb{A}_C \mathsf{m}_C \star \mathbb{B}_C$. Applying the same rotation to every kernel gives $\check{C} = \check{A} \star \check{B}$. Only stationarity reduces these operations to frequency products.

$$\begin{aligned}\begin{pmatrix} G^R & G^K \\ 0 & G^A \end{pmatrix} = \begin{pmatrix} g^R & g^K \\ 0 & g^A \end{pmatrix} \\ + \begin{pmatrix} g^R & g^K \\ 0 & g^A \end{pmatrix} \star \begin{pmatrix} \Sigma^R & \Sigma^K \\ 0 & \Sigma^A \end{pmatrix} \star \begin{pmatrix} G^R & G^K \\ 0 & G^A \end{pmatrix}.\end{aligned} \tag{2.2.41}$$

$$\begin{aligned}&\check{g} \star \check{\Sigma} \star \check{G} \\ &= \begin{pmatrix} g^R \star \Sigma^R & g^R \star \Sigma^K + g^K \star \Sigma^A \\ 0 & g^A \star \Sigma^A \end{pmatrix} \star \begin{pmatrix} G^R & G^K \\ 0 & G^A \end{pmatrix}.\end{aligned} \tag{2.2.42}$$

$$\begin{aligned}&\check{g} \star \check{\Sigma} \star \check{G} \\ &= \begin{pmatrix} g^R \star \Sigma^R \star G^R & g^R \star \Sigma^R \star G^K \\ & + (g^R \star \Sigma^K + g^K \star \Sigma^A) \star G^A \\ 0 & g^A \star \Sigma^A \star G^A \end{pmatrix}.\end{aligned} \tag{2.2.43}$$

$$C = A \circ_C B, \qquad C(1,1') = \int_C d2\, A(1,2) B(2,1'). \tag{2.2.44}$$

$$(A \circ_C B)^\lessgtr = A^R \star B^\lessgtr + A^\lessgtr \star B^A. \tag{2.2.45}$$

$$(A \circ_C B)^{R,A} = A^{R,A} \star B^{R,A}. \tag{2.2.46}$$

$$\begin{aligned}(A \circ_C B \circ_C C)^\lessgtr = A^R \star B^R \star C^\lessgtr + A^R \star B^\lessgtr \star C^A \\ + A^\lessgtr \star B^A \star C^A.\end{aligned} \tag{2.2.47}$$

$$(A \circ_C B \circ_C C)^{R,A} = A^{R,A} \star B^{R,A} \star C^{R,A}. \tag{2.2.48}$$

These are the real-branch Langreth rules. An imaginary preparation branch would add mixed-contour terms. In a lesser product, one factor carries occupied information while factors to its left are retarded and factors to its right are advanced. This ordered structure must be retained for multiorbital matrices.

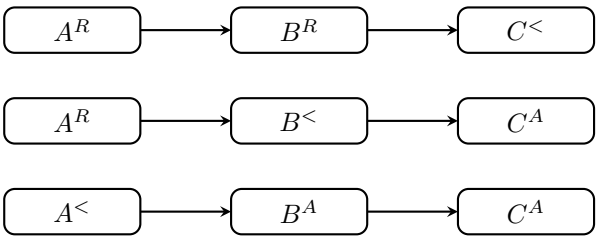


Figure 2.14: The three ordered contributions to a lesser triple product. The occupied component can occur in each of the three positions; retarded factors precede it and advanced factors follow it.

$$G^R = g^R + g^R \star \Sigma^R \star G^R. \tag{2.2.49}$$

$$G^A = g^A + g^A \star \Sigma^A \star G^A. \tag{2.2.50}$$

$$G^K = g^K + g^R \star \Sigma^R \star G^K + g^R \star \Sigma^K \star G^A + g^K \star \Sigma^A \star G^A. \tag{2.2.51}$$

$$G^K = G^R - G^A + 2G^<. \tag{2.2.52}$$

$$\begin{aligned} G^< = (1 + G^R \star \Sigma^R) \star g^< \star (1 + \Sigma^A \star G^A) \\ + G^R \star \Sigma^< \star G^A. \end{aligned} \tag{2.2.53}$$

To see where its first term comes from, apply the triple-product rule to Dyson's equation:

$$G^< = g^< + g^R \star \Sigma^R \star G^< + g^R \star \Sigma^< \star G^A + g^< \star \Sigma^A \star G^A, \tag{2.2.54}$$

$$(1 - g^R \star \Sigma^R) \star G^< = g^< \star (1 + \Sigma^A \star G^A) + g^R \star \Sigma^< \star G^A. \tag{2.2.55}$$

The retarded Dyson equation implies

$$(1 - g^R \star \Sigma^R)^{-1} = 1 + G^R \star \Sigma^R, \qquad (1 + G^R \star \Sigma^R) \star g^R = G^R. \tag{2.2.56}$$

Multiplication on the left yields (2.2.53). The identity 1 is an identity kernel. If $\Sigma^< = 0$ for static hopping, this first term still supplies the reservoir occupations; $G^<$ is generally nonzero.

When both the reference functions and the perturbation kernel are known, these equations determine all components. For an interacting problem, determining the self-energy is an additional problem.

A convenient strategy is to choose g for equilibrated reference subsystems, whose spectral and occupation functions are available independently.

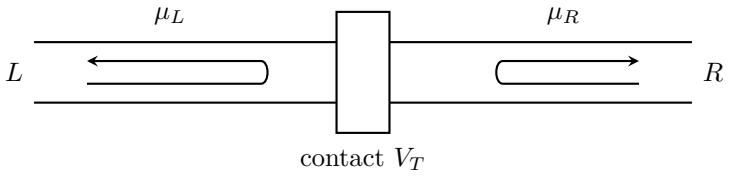


Figure 2.15: Two free reservoirs joined at a local barrier. Their chemical potentials set the incident occupations. The contact Hamiltonian transfers particles between the endpoints; the reservoir labels do not imply spin or a second independent channel.

2.3 A single-barrier contact

2.3.1 Hamiltonian, reservoir preparation, and current

$$H_L = \sum_k \epsilon_{Lk} c_{Lk}^\dagger c_{Lk}. \tag{2.3.1}$$

$$H_R = \sum_k \epsilon_{Rk} c_{Rk}^\dagger c_{Rk}. \tag{2.3.2}$$

$$H_T = \sum_{kq} \left(V_{kq} c_{Lk}^\dagger c_{Rq} + V_{kq}^* c_{Rq}^\dagger c_{Lk} \right). \tag{2.3.3}$$

$$V_{kq} = \frac{V_T}{L_{\mathrm{sys}}} \qquad \text{for equal normalization lengths } L_{\mathrm{sys}}. \tag{2.3.4}$$

For one independent fermionic channel, define

$$\psi_\ell(x) = \frac{1}{\sqrt{L_{\mathrm{sys}}}} \sum_k e^{ikx} c_{\ell k}, \qquad \ell = L, R, \tag{2.3.5}$$

so that ψ_ℓ has dimension length$^{-1/2}$. The local amplitude V_T then has dimension energy times length, whereas V_{kq} has dimension energy. The continuum limit is $L_{\mathrm{sys}}^{-1} \sum_k \to \int dk/(2\pi)$.

$$H_T = V_T \psi_L^\dagger(0) \psi_R(0) + V_T^* \psi_R^\dagger(0) \psi_L(0). \tag{2.3.6}$$

$$N_L = \sum_k c_{Lk}^\dagger c_{Lk}. \tag{2.3.7}$$

$$I(t) = -e \frac{dN_L(t)}{dt}, \qquad e > 0. \tag{2.3.8}$$

$$I(t) = -ie[H, N_L](t). \tag{2.3.9}$$

$$I(t) = -ie[H_T, N_L](t), \qquad [H_L + H_R, N_L] = 0. \tag{2.3.10}$$

The canonical anticommutators imply $[N_L, c_{Lk}^\dagger] = c_{Lk}^\dagger$ and $[N_L, c_{Lk}] = -c_{Lk}$, while N_L commutes with right-reservoir bilinears. Therefore

$$[c_{Lk}^\dagger c_{Rq}, N_L] = -c_{Lk}^\dagger c_{Rq}, \qquad [c_{Rq}^\dagger c_{Lk}, N_L] = +c_{Rq}^\dagger c_{Lk}. \tag{2.3.11}$$

$$I(t) = \frac{ie}{L_{\mathrm{sys}}} \sum_{kq} \left[V_T c_{Lk}^\dagger(t) c_{Rq}(t) - V_T^* c_{Rq}^\dagger(t) c_{Lk}(t) \right]. \tag{2.3.12}$$

$$I(t) = ie\left[V_T\psi_L^\dagger(0,t)\psi_R(0,t) - V_T^*\psi_R^\dagger(0,t)\psi_L(0,t)\right]. \tag{2.3.13}$$

The current operator is Hermitian. Positive current corresponds to electron number leaving L for R. Since $Q_L = -eN_L$, it also equals $\dot{Q}_L$; the signed electron-charge flux oriented from L to R is $-I$. Directly, $[H_T, N_L + N_R] = 0$, so $\dot{N}_L + \dot{N}_R = 0$. No reservoir is a source or sink of total particles in this closed contact Hamiltonian.

Why do the real-time lead Hamiltonians contain $\epsilon_{\ell k}$, rather than $\epsilon_{\ell k} - \mu_\ell$? Thermal preparation and real-time evolution are different operations.

When $\mu_L \neq \mu_R$, one common energy shift cannot remove both reservoir chemical potentials. Separate time-dependent phase transformations can shift both lead spectra, but then the tunnelling term acquires the relative phase set by $\mu_L - \mu_R$.

The chemical potentials enter the initial occupations used to evaluate correlations.

$$\rho_0 = \frac{1}{Z_0}\exp\left[-\beta\sum_{\ell=L,R}(H_\ell - \mu_\ell N_\ell)\right]. \tag{2.3.14}$$

This is a product preparation of independently equilibrated leads. It is not a Gibbs state of the coupled biased device. In the physical-energy representation used here, $f_\ell(\omega) = [e^{\beta(\omega-\mu_\ell)} + 1]^{-1}$, whereas real-time free evolution is generated by $H_L + H_R$. The later gauge transformation changes these two descriptions together.

$$\langle I(t)\rangle = e\left[V_T G^<_{RL}(0,t;0,t) - V_T^* G^<_{LR}(0,t;0,t)\right]. \tag{2.3.15}$$

The first subscript of $G^<_{ij}$ always labels the annihilation field:

$$G^<_{RL}(0,t;0,t) = i\langle\psi_L^\dagger(0,t)\psi_R(0,t)\rangle. \tag{2.3.16}$$

Its two position-time arguments belong to R and L, respectively.

$$G^{\lessgtr}_{\ell\ell'}(t,t') \equiv G^{\lessgtr}_{\ell\ell'}(0,t;0,t'). \tag{2.3.17}$$

$$I = e\int\frac{d\omega}{2\pi}\left[V_T G^<_{RL}(\omega) - V_T^* G^<_{LR}(\omega)\right]. \tag{2.3.18}$$

The last expression assumes stationarity. It shows precisely which mixed lesser functions must be calculated. The relation $[G^<_{RL}(\omega)]^* = -G^<_{LR}(\omega)$ ensures that the displayed current is real.

$$\check{\Sigma}_V = \begin{pmatrix}\mathsf{V} & 0\\ 0 & \mathsf{V}\end{pmatrix}, \qquad \mathsf{V} = \begin{pmatrix}0 & V_T\\ V_T^* & 0\end{pmatrix}_{L,R}. \tag{2.3.19}$$

Each instantaneous tunnelling vertex joins $\psi_L^\dagger(0,t^\kappa)$ and $\psi_R(0,t^\kappa)$ on the *same* branch. Its retarded and advanced kernels are $\Sigma_V^R(t,t') = \Sigma_V^A(t,t') = \mathsf{V}\delta(t-t')$; its lesser, greater, and Keldysh kernels vanish.

The outer triangular matrix is in the rotated contour space. Each of its entries is a separate matrix in lead space. An off-diagonal element in lead space describes transfer between reservoirs; a lesser component in contour space describes occupied correlation. These meanings are independent.

$$G^< = (1 + G^R\mathsf{V})g^<(1 + \mathsf{V}G^A) + G^R\underbrace{\Sigma_V^<}_{0}G^A. \tag{2.3.20}$$

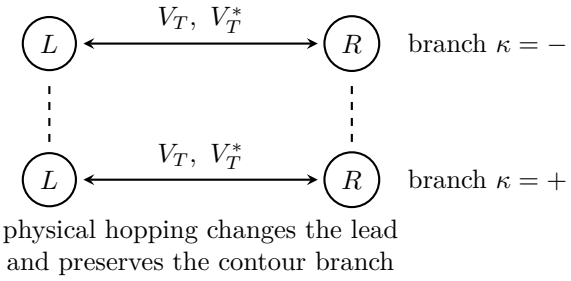


Figure 2.16: Independent index spaces. A tunnelling vertex changes the lead index while retaining its contour branch. The rotated contour kernel has equal retarded and advanced lead-space matrices and a zero Keldysh entry.

$$\begin{aligned} G^<_{RL} = \underbrace{g^<_{RL}}_{0} &+ G^R_{RR} V_T^* g^<_L + G^R_{RL} V_T \underbrace{g^<_{RL}}_{0} \\ &+ g^<_R V_T^* G^A_{LL} + G^R_{RR} V_T^* g^<_L V_T G^A_{RL} \\ &+ G^R_{RL} V_T g^<_R V_T^* G^A_{LL}. \end{aligned} \tag{2.3.21}$$

The uncoupled reference has $g_{RL} = g_{LR} = 0$, which removes the explicitly marked mixed-reference terms. The remaining terms show how occupied states from each reservoir propagate into a mixed contact correlation. The zero lesser hopping kernel has not removed those occupations.

2.3.2 Solving the coupled contact Green's functions

$$G^R_{RR} = g^R_R + g^R_R V_T^* G^R_{LR}. \tag{2.3.22}$$

$$G^R_{LR} = \underbrace{g^R_{LR}}_{0} + g^R_L V_T G^R_{RR}. \tag{2.3.23}$$

$$G^R_{RR} = g^R_R + g^R_R V_T^* g^R_L V_T G^R_{RR}. \tag{2.3.24}$$

$$G^R_{RR} = \frac{g^R_R}{1 - g^R_R V_T^* g^R_L V_T}. \tag{2.3.25}$$

$$G^R_{LR} = \frac{g^R_L V_T g^R_R}{1 - g^R_R V_T^* g^R_L V_T}. \tag{2.3.26}$$

$$G^R_{RR} = \frac{g^R_R}{1 - |V_T|^2 g^R_R g^R_L}. \tag{2.3.27}$$

$$G^R_{LR} = \frac{V_T g^R_L g^R_R}{1 - |V_T|^2 g^R_R g^R_L}. \tag{2.3.28}$$

$$G^R_{LL} = \frac{g^R_L}{1 - |V_T|^2 g^R_L g^R_R}. \tag{2.3.29}$$

$$G^R_{RL} = \frac{V_T^* g^R_L g^R_R}{1 - |V_T|^2 g^R_L g^R_R}. \tag{2.3.30}$$

Define $D^R = 1 - |V_T|^2 g_L^R g_R^R$. The factor $|V_T|^2 g_R^R g_L^R$ is one return excursion from R to L and back. The Dyson solution sums repeated excursions: $g_R^R[1 + x + x^2 + \cdots] = g_R^R/(1-x)$ when the series converges, followed by analytic continuation with the retarded prescription. The scalar rearrangements above apply to one contact channel. In an orbital matrix problem one must retain ordered matrix inverses, such as $(1 - g_R^R \mathsf{V}_{RL} g_L^R \mathsf{V}_{LR})^{-1} g_R^R$ with $\mathsf{V}_{RL} = \mathsf{V}_{LR}^\dagger$, and cannot commute factors.

The advanced functions follow from $G^A = (G^R)^\dagger$ at real frequency, with $D^A = (D^R)^*$.

$$G_{LL}^A = \frac{g_L^A}{1 - |V_T|^2 g_L^A g_R^A}. \tag{2.3.31}$$

$$G_{RR}^A = \frac{g_R^A}{1 - |V_T|^2 g_L^A g_R^A}. \tag{2.3.32}$$

$$G_{LR}^A = \frac{V_T g_L^A g_R^A}{1 - |V_T|^2 g_L^A g_R^A}. \tag{2.3.33}$$

$$G_{RL}^A = \frac{V_T^* g_L^A g_R^A}{1 - |V_T|^2 g_L^A g_R^A}. \tag{2.3.34}$$

$$G_{RL}^< = G_{RR}^R V_T^* g_L^< (1 + V_T G_{RL}^A) + (1 + G_{RL}^R V_T) g_R^< V_T^* G_{LL}^A. \tag{2.3.35}$$

$$\begin{aligned} G_{RL}^< = {} & \frac{V_T^* g_R^R g_L^<}{1 - |V_T|^2 g_R^R g_L^R} \left(1 + \frac{|V_T|^2 g_L^A g_R^A}{1 - |V_T|^2 g_L^A g_R^A}\right) \\ & + \left(1 + \frac{|V_T|^2 g_L^R g_R^R}{1 - |V_T|^2 g_L^R g_R^R}\right) \frac{V_T^* g_R^< g_L^A}{1 - |V_T|^2 g_L^A g_R^A}. \end{aligned} \tag{2.3.36}$$

Each parenthesis is $1 + x/(1-x) = 1/(1-x)$. Consequently, the first term has denominator $D^R D^A$ and numerator $V_T^* g_R^R g_L^<$; the second has the same denominator and numerator $V_T^* g_R^< g_L^A$. No occupation terms are discarded in making this common denominator.

$$G_{RL}^< = \frac{V_T^* (g_R^R g_L^< + g_R^< g_L^A)}{(1 - |V_T|^2 g_R^R g_L^R)(1 - |V_T|^2 g_L^A g_R^A)}. \tag{2.3.37}$$

$$G_{LR}^< = \frac{V_T (g_L^< g_R^A + g_L^R g_R^<)}{(1 - |V_T|^2 g_R^R g_L^R)(1 - |V_T|^2 g_L^A g_R^A)}. \tag{2.3.38}$$

The two numerator terms identify occupation injected from L and from R, dressed by propagation through the contact. Since $D^R D^A = |D^R|^2$, the denominator is real and nonnegative. The mixed correlators vanish as $V_T \to 0$ and satisfy $[G_{RL}^<]^* = -G_{LR}^<$, providing independent checks of the conjugate amplitude and the index order.

2.3.3 From mixed occupations to a current integral

$$I = e \left[V_T G_{RL}^<(t,t) - V_T^* G_{LR}^<(t,t) \right]. \tag{2.3.39}$$

$$I = e \int \frac{d\omega}{2\pi} \left[V_T G_{RL}^<(\omega) - V_T^* G_{LR}^<(\omega) \right]. \tag{2.3.40}$$

$$I = e|V_T|^2 \int \frac{d\omega}{2\pi} \frac{g_R^R g_L^< + g_R^< g_L^A - g_L^< g_R^A - g_L^R g_R^<}{(1 - |V_T|^2 g_L^R g_R^R)(1 - |V_T|^2 g_L^A g_R^A)}. \tag{2.3.41}$$

$$g_\ell^<(\omega) = f_\ell(\omega)\left[g_\ell^A(\omega) - g_\ell^R(\omega)\right], \qquad \ell = L, R. \tag{2.3.42}$$

This fluctuation–occupation relation applies separately to each disconnected equilibrium lead. Write the numerator of (2.3.41) as $\mathcal{N}$. Because the contact functions are scalar,

$$\begin{aligned}\mathcal{N} &= g_L^<(g_R^R - g_R^A) + g_R^<(g_L^A - g_L^R) && (2.3.43)\\ &= f_L(g_L^A - g_L^R)(g_R^R - g_R^A) + f_R(g_R^A - g_R^R)(g_L^A - g_L^R) && (2.3.44)\\ &= (g_L^A - g_L^R)(g_R^R - g_R^A)(f_L - f_R). && (2.3.45)\end{aligned}$$

The sign in the second term changes because $g_R^A - g_R^R = -(g_R^R - g_R^A)$.

$$I = e\int \frac{d\omega}{2\pi}\frac{|V_T|^2(g_L^A - g_L^R)(g_R^R - g_R^A)}{(1 - |V_T|^2 g_L^R g_R^R)(1 - |V_T|^2 g_L^A g_R^A)}\,[f_L - f_R]. \tag{2.3.46}$$

$$\mathcal{T}(\omega) \equiv \frac{|V_T|^2(g_L^A - g_L^R)(g_R^R - g_R^A)}{|1 - |V_T|^2 g_L^R g_R^R|^2}, \tag{2.3.47}$$

$$I = e\int \frac{d\omega}{2\pi}\mathcal{T}(\omega)[f_L - f_R]. \tag{2.3.48}$$

The denominator arises from repeated propagation through the contact. Like the dressed denominator in the impurity example, it reorganizes the poles and spectral weight through coupling to a continuum. In this barrier model its later wide-band parameter is a dimensionless contact strength; it is not the impurity energy half-width Γ_d.

For passive normal leads, $A_\ell = i(g_\ell^R - g_\ell^A) = 2\pi\nu_\ell \geq 0$. Hence $(g_L^A - g_L^R)(g_R^R - g_R^A) = (iA_L)(-iA_R) = A_L A_R$. The result can be written

$$\mathcal{T}(E) = \frac{4\pi^2|V_T|^2\nu_L(E)\nu_R(E)}{|1 - |V_T|^2 g_L^R(E) g_R^R(E)|^2}, \tag{2.3.49}$$

$$I = \frac{e}{h}\int dE\,\mathcal{T}(E)[f_L(E) - f_R(E)]. \tag{2.3.50}$$

The restored measure is dE/h, with $h = 2\pi\hbar$. A channel contributes only where the two incident occupations differ. Equal chemical potentials and temperatures give zero current; exchanging otherwise identical reservoirs reverses its sign. The next step is to calculate the local free-lead functions that enter this transmission.

2.3.4 Equilibrium contact functions for linear dispersion

Set the contact hopping to zero, $V_T = 0$. Each disconnected lead is then an equilibrium channel, and its functions g_ℓ, $\ell = L, R$, can be evaluated independently. We use the physical Hamiltonian for real-time evolution and the chemical potential only in the thermal weight:

$$H_\ell = \sum_k \epsilon_{\ell k} c_{\ell k}^\dagger c_{\ell k}, \qquad f_\ell(E) = \frac{1}{e^{\beta(E-\mu_\ell)} + 1}, \qquad c_{\ell k}(t) = e^{-i\epsilon_{\ell k}t} c_{\ell k}. \tag{2.3.51}$$

The lead labels identify reservoirs. They do not introduce a second spin channel or a second propagation branch.

Greater function.

$$g^{>}_{\ell k}(\omega) = \int_{-\infty}^{\infty} dt\, e^{i\omega t} g^{>}_{\ell k}(t), \qquad g^{>}_{\ell k}(t) = -i[1 - f_\ell(\epsilon_{\ell k})] e^{-i\epsilon_{\ell k} t}. \tag{2.3.52}$$

The chemical potential enters through $f_\ell(\epsilon_{\ell k})$.

$$\begin{aligned} g^{>}_{\ell k}(\omega) &= \int_{-\infty}^{\infty} dt\, e^{i(\omega - \epsilon_{\ell k})t} \{-i[1 - f_\ell(\epsilon_{\ell k})]\} \\ &= -2\pi i [1 - f_\ell(\epsilon_{\ell k})] \delta(\omega - \epsilon_{\ell k}). \end{aligned} \tag{2.3.53}$$

The Fourier integral equals $2\pi\delta(\omega - \epsilon_{\ell k})$ as a distribution. Its coefficient is the probability that the mode is empty, as required by $\langle c_{\ell k} c^\dagger_{\ell k} \rangle = 1 - f_\ell$.

Lesser function.

$$g^{<}_{\ell k}(\omega) = \int_{-\infty}^{\infty} dt\, e^{i\omega t} g^{<}_{\ell k}(t) = 2\pi i f_\ell(\epsilon_{\ell k}) \delta(\omega - \epsilon_{\ell k}). \tag{2.3.54}$$

Here $g^{<}_{\ell k}(t) = i f_\ell(\epsilon_{\ell k}) e^{-i\epsilon_{\ell k} t}$. The occupation appears in the lesser function, while the greater function counts available holes.

Retarded and advanced functions.

$$g^{R}_{\ell k}(\omega) = -i \int_{-\infty}^{\infty} dt\, e^{i\omega t} \Theta(t) e^{-i\epsilon_{\ell k} t} = \frac{1}{\omega - \epsilon_{\ell k} + i0^+}. \tag{2.3.55}$$

To evaluate the integral, insert $e^{-\eta_0 t}$ for $t > 0$, with $\eta_0 > 0$, and then take $\eta_0 \to 0^+$:

$$-i \int_0^{\infty} dt\, e^{[i(\omega - \epsilon_{\ell k}) - \eta_0]t} = \frac{-i}{\eta_0 - i(\omega - \epsilon_{\ell k})} = \frac{1}{\omega - \epsilon_{\ell k} + i\eta_0}. \tag{2.3.56}$$

$$g^{A}_{\ell k}(\omega) = \frac{1}{\omega - \epsilon_{\ell k} - i0^+}. \tag{2.3.57}$$

The infinitesimal enforces a boundary condition. It is not a physical linewidth. For these quadratic modes, the retarded function is independent of the occupation because $\{c_{\ell k}, c^\dagger_{\ell k}\} = 1$.

From momentum space to the local contact. For normalization length L_{sys}, define

$$\psi_\ell(x) = \frac{1}{\sqrt{L_{\text{sys}}}} \sum_k e^{ikx} c_{\ell k}, \qquad g_\ell(\omega) = g_\ell(0, 0; \omega) = \frac{1}{L_{\text{sys}}} \sum_k g_{\ell k}(\omega). \tag{2.3.58}$$

Translation invariance makes the momentum correlator diagonal. In the continuum limit $L_{\text{sys}}^{-1} \sum_k \to \int dk/(2\pi)$.

$$\begin{aligned} g^{>}_{\ell}(\omega) &= \int \frac{dk}{2\pi}\, g^{>}_{\ell k}(\omega) \\ &= -i \int dk\, [1 - f_\ell(\epsilon_{\ell k})] \delta(\omega - \epsilon_{\ell k}). \end{aligned} \tag{2.3.59}$$

$$g^{>}_{\ell}(\omega) = -i \int d\epsilon \left| \frac{d\epsilon_{\ell k}}{dk} \right|^{-1} [1 - f_\ell(\epsilon)] \delta(\omega - \epsilon). \tag{2.3.60}$$

This expression describes one monotone branch. A decreasing dispersion reverses the integration endpoints; the absolute value keeps the density of states positive. Several roots of $E = \epsilon_{\ell k}$ contribute a sum of inverse absolute velocities:

$$\nu_\ell(E) = \int \frac{dk}{2\pi} \delta(E - \epsilon_{\ell k}) = \sum_{k_j :\, \epsilon_{\ell k_j} = E} \frac{1}{2\pi |\partial_k \epsilon_{\ell k_j}|}. \tag{2.3.61}$$

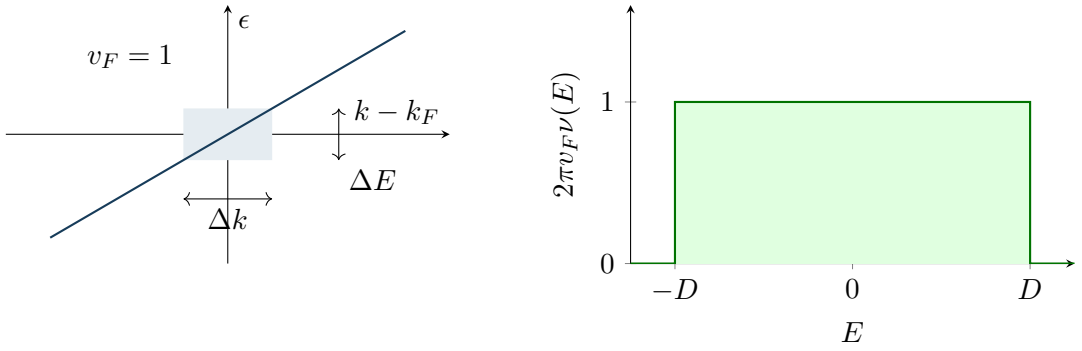


Figure 2.17: One linear propagation branch has constant density of states. The left panel uses $\epsilon = v_F(k - k_F)$, with $v_F = 1$ and $k_F = 0$ in illustrative units; the shaded energy interval of width ΔE corresponds to $\Delta k = \Delta E/v_F$. The right panel shows $\nu(E) = 1/(2\pi v_F)$ inside the illustrative cutoff $|E| < D = 2$. The band edges regulate the local field; the formulas used in transport apply away from them.

For one linear branch of speed $v_F > 0$,

$$\frac{1}{v_F} = 2\pi\nu_0. \tag{2.3.62}$$

$$g_\ell^>(\omega) = -2\pi i\nu_0[1 - f_\ell(\omega)]. \tag{2.3.63}$$

$$g_\ell^<(\omega) = 2\pi i\nu_0 f_\ell(\omega). \tag{2.3.64}$$

$$g_\ell^R(\omega) = \int \frac{dk}{2\pi} g_{\ell k}^R(\omega) = \frac{1}{2\pi}\int dk\, \frac{1}{\omega - \epsilon_{\ell k} + i0^+}. \tag{2.3.65}$$

$$g_\ell^R(\omega) = \nu_0 \int d\epsilon\, \frac{1}{\omega - \epsilon + i0^+} \simeq -i\pi\nu_0. \tag{2.3.66}$$

$$g_\ell^A(\omega) \simeq +i\pi\nu_0. \tag{2.3.67}$$

The last step is a symmetric wide-band approximation. For a constant density of states supported on $[-D, D]$, the explicit integral is

$$g_\ell^R(\omega) = \nu_0 \ln\left|\frac{\omega + D}{\omega - D}\right| - i\pi\nu_0\Theta(D - |\omega|). \tag{2.3.68}$$

At $|\omega| \ll D$, the real part is $2\nu_0\omega/D + O((\omega/D)^3)$. It therefore vanishes in the symmetric wide-band limit at fixed ω. For an asymmetric band, its principal-value contribution must be retained or absorbed into a specified energy reference. With the continuum field normalization, ν_0 has units $(\text{energy} \times \text{length})^{-1}$ and V_T has units energy times length. The local spectral density $A_\ell = 2\pi\nu_0$ is not a unit-normalized single-mode pole.

2.3.5 Transmission and conductance

The local lead functions now complete the contact-current calculation. Choose the relative phase of the two normal lead fields so that V_T is real and nonnegative. For a single contact this removes its phase without changing an observable. The next explicit Wick expansion uses this phase choice. For two identical wide-band leads, $g_L^R = g_R^R = -i\pi\nu_0$ and $g_L^A = g_R^A = i\pi\nu_0$.

$$\lambda_c = \pi^2 V_T^2 \nu_0^2. \tag{2.3.69}$$

For unequal local densities of states, the corresponding definition is $\lambda_c = \pi^2|V_T|^2\nu_L\nu_R$. In either normalization it is dimensionless. It is distinct from the impurity energy half-width Γ_d. Substitution into the mixed lesser functions gives

$$\begin{aligned} G^<_{RL} &= \frac{V_T[(-i\pi\nu_0)(2\pi i\nu_0 f_L) + (2\pi i\nu_0 f_R)(i\pi\nu_0)]}{(1+\lambda_c)^2} \\ &= \frac{2\pi^2\nu_0^2 V_T}{(1+\lambda_c)^2}(f_L - f_R), \qquad G^<_{LR} = -G^<_{RL}. \end{aligned} \tag{2.3.70}$$

Thus $eV_T(G^<_{RL} - G^<_{LR}) = e\,4\lambda_c(f_L - f_R)/(1+\lambda_c)^2$.

$$I = \frac{e}{2\pi}\frac{4\lambda_c}{(1+\lambda_c)^2}\int_{-\infty}^{\infty} d\omega\,[f_L(\omega) - f_R(\omega)]. \tag{2.3.71}$$

The coefficient is the one-channel transmission probability in the Landauer–Büttiker expression:

$$\mathcal{T} = \frac{4\lambda_c}{(1+\lambda_c)^2}. \tag{2.3.72}$$

$$\int_{-\infty}^{\infty} d\omega\,[f_L(\omega) - f_R(\omega)] = \mu_L - \mu_R = \Delta\mu. \tag{2.3.73}$$

This identity assumes equal reservoir temperatures and an effectively unbounded flat band. To verify it, differentiate the convergent difference integral with respect to μ_L:

$$\frac{\partial}{\partial\mu_L}\int d\omega(f_L - f_R) = -\int d\omega\,\partial_\omega f_L = 1. \tag{2.3.74}$$

Its value vanishes at $\mu_L = \mu_R$, fixing the integration constant. At zero temperature the same result is the width of the occupied-to-empty transport window. Restoring units gives $I = (e/h)\mathcal{T}\Delta\mu$. The convention $I = -e\dot{N}_L$, with $e > 0$, makes $I > 0$ when electron number leaves L for R. It also gives $I = \dot{Q}_L$, where $Q_L = -eN_L$.

$$\mathcal{G} = e\frac{\partial I}{\partial(\mu_L - \mu_R)} = \frac{e^2}{2\pi\hbar}\frac{4\lambda_c}{(1+\lambda_c)^2} = \frac{e^2}{h}\mathcal{T}. \tag{2.3.75}$$

Here $\mathcal{T}$ denotes transmission, while $\mathcal{G}$ denotes conductance. The voltage dictionary is $V_{\text{bias}} = \Delta\mu/e$. No spin factor has been included. More generally, linear response gives

$$\mathcal{G} = \frac{e^2}{h}\int dE\,\mathcal{T}(E)[-\partial_E f(E)], \tag{2.3.76}$$

which approaches $(e^2/h)\mathcal{T}(E_F)$ at zero temperature. The bound and the matching point follow without a scattering ansatz:

$$1 - \mathcal{T} = \frac{(1-\lambda_c)^2}{(1+\lambda_c)^2} \geq 0, \qquad \mathcal{T}(1) = 1, \qquad \mathcal{T} = 4\lambda_c + O(\lambda_c^2). \tag{2.3.77}$$

The invariance $\mathcal{T}(\lambda_c) = \mathcal{T}(1/\lambda_c)$ belongs to this local contact parametrization. A large contact hopping is therefore not universally equivalent to a small microscopic barrier height.

2.3.6 Connected tunnelling-current noise and Wick contractions

$$C_{II}(t,0) = \langle\delta I(t)\delta I(0)\rangle, \qquad \delta I(t) = I(t) - \langle I(t)\rangle. \tag{2.3.78}$$

This is an ordered connected correlator. Before doing the algebra, fix the spectral normalization:

$$\mathcal{S}^{\text{sym}}(\Omega) = \frac{1}{2}\int_{-\infty}^{\infty} dt\,e^{i\Omega t}\langle\{\delta I(t), \delta I(0)\}\rangle, \tag{2.3.79}$$

$$\mathcal{S}^{(2)}(\Omega) = 2\mathcal{S}^{\text{sym}}(\Omega), \qquad \mathcal{S}^{(2)}(0) = 2\int_{-\infty}^{\infty} dt\,C_{II}(t,0). \tag{2.3.80}$$

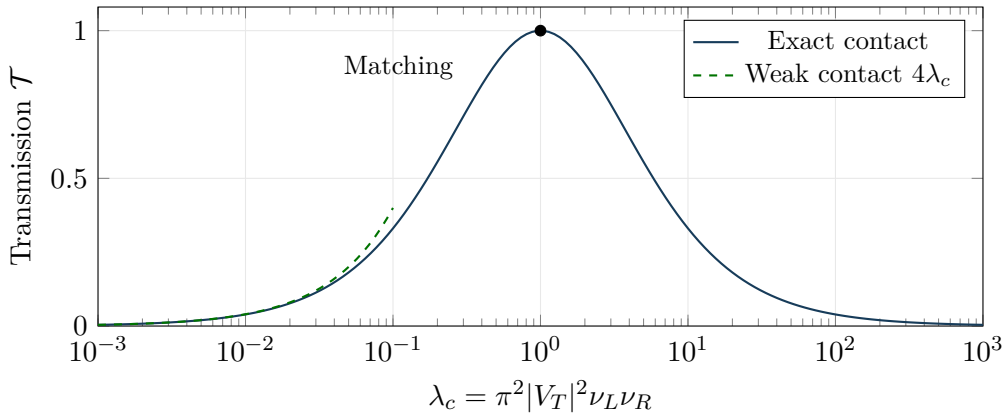


Figure 2.18: Analytic transmission of the symmetric wide-band contact, $\mathcal{T} = 4\lambda_c/(1+\lambda_c)^2$, with $\lambda_c \in [10^{-3}, 10^3]$. The dashed curve is the weak-contact approximation $\mathcal{T}_0 = 4\lambda_c$, shown only for $\lambda_c \le 0.1$. The matching point $\lambda_c = 1$ has unit transmission; reciprocal contact strengths have identical transmission. This is a model curve, not measured data.

The last identity requires stationarity and zero measurement frequency. The factor 2 belongs to the doubled spectrum, not to (2.3.78).

The contact position $x = 0$ is suppressed in the following fields. For readability write $\psi_{\ell t} \equiv \psi_\ell(0,t)$. With the real hopping chosen above,

$$I(t) = ieV_T(\psi^\dagger_{Lt}\psi_{Rt} - \psi^\dagger_{Rt}\psi_{Lt}). \tag{2.3.81}$$

Expanding $\langle I(t)I(0)\rangle - \langle I(t)\rangle\langle I(0)\rangle$ gives all four ordered four-point functions and the disconnected subtraction:

$$\begin{aligned}\frac{C_{II}(t,0)}{e^2V_T^2} &= \langle\psi^\dagger_{Lt}\psi_{Rt}\psi^\dagger_{R0}\psi_{L0}\rangle \\ &\quad + \langle\psi^\dagger_{Rt}\psi_{Lt}\psi^\dagger_{L0}\psi_{R0}\rangle \\ &\quad - \langle\psi^\dagger_{Lt}\psi_{Rt}\psi^\dagger_{L0}\psi_{R0}\rangle \\ &\quad - \langle\psi^\dagger_{Rt}\psi_{Lt}\psi^\dagger_{R0}\psi_{L0}\rangle \\ &\quad + \big[\langle\psi^\dagger_{Lt}\psi_{Rt}\rangle - \langle\psi^\dagger_{Rt}\psi_{Lt}\rangle\big]\big[\langle\psi^\dagger_{L0}\psi_{R0}\rangle - \langle\psi^\dagger_{R0}\psi_{L0}\rangle\big].\end{aligned} \tag{2.3.82}$$

The plus sign of the final product is fixed by $i^2 = -1$ in the mean current product and the minus sign of the connected subtraction. Dropping this term before factorization would retain spurious mean-current noise.

For the first four-point function one can distinguish the two real-time orderings explicitly:

$$\langle\psi^\dagger_{Lt}\psi_{Rt}\psi^\dagger_{R0}\psi_{L0}\rangle = \begin{cases}\langle \mathsf{T}\{[\psi^\dagger_{Lt}\psi_{Rt}][\psi^\dagger_{R0}\psi_{L0}]\}\rangle, & t > 0,\\ \langle \overline{\mathsf{T}}\{[\psi^\dagger_{Lt}\psi_{Rt}][\psi^\dagger_{R0}\psi_{L0}]\}\rangle, & t < 0.\end{cases} \tag{2.3.83}$$

The ordering inside each equal-time bilinear is kept as displayed; equivalently one can use a fixed point-splitting prescription. Exchanging two bilinears exchanges four fermion pairs and produces no overall sign. The actual contraction below is the ordinary ordered Gaussian expectation, so no special value of $\Theta(0)$ enters. Wick's theorem applies to either ordering. Its required premise here is a normal Gaussian state, obtained from independent Gaussian leads by evolution with the quadratic number-conserving contact Hamiltonian. Anomalous contractions vanish. With arbitrary normal fermions,

$$\langle a^\dagger b c^\dagger d\rangle = \langle a^\dagger b\rangle\langle c^\dagger d\rangle - \underbrace{\langle a^\dagger c^\dagger\rangle\langle bd\rangle}_{0} + \langle a^\dagger d\rangle\langle bc^\dagger\rangle. \tag{2.3.84}$$

The surviving exchange pairing has a plus sign in this operator order. For an interacting non-Gaussian state, an independent connected four-point vertex would remain.

$$\langle \psi^\dagger_{Lt}\psi_{Rt}\psi^\dagger_{R0}\psi_{L0}\rangle = \langle \psi^\dagger_{Lt}\psi_{Rt}\rangle\langle \psi^\dagger_{R0}\psi_{L0}\rangle + \langle \psi^\dagger_{Lt}\psi_{L0}\rangle\langle \psi_{Rt}\psi^\dagger_{R0}\rangle. \tag{2.3.85}$$

Applying the same rule to the other three functions and expanding the last product of (2.3.82) gives the complete twelve-term expression:

$$\begin{aligned}\frac{C_{II}(t,0)}{e^2V_T^2} = {}& \langle \psi^\dagger_{Lt}\psi_{Rt}\rangle\langle \psi^\dagger_{R0}\psi_{L0}\rangle + \langle \psi^\dagger_{Lt}\psi_{L0}\rangle\langle \psi_{Rt}\psi^\dagger_{R0}\rangle \\ &+ \langle \psi^\dagger_{Rt}\psi_{Lt}\rangle\langle \psi^\dagger_{L0}\psi_{R0}\rangle + \langle \psi^\dagger_{Rt}\psi_{R0}\rangle\langle \psi_{Lt}\psi^\dagger_{L0}\rangle \\ &- \langle \psi^\dagger_{Lt}\psi_{Rt}\rangle\langle \psi^\dagger_{L0}\psi_{R0}\rangle - \langle \psi^\dagger_{Lt}\psi_{R0}\rangle\langle \psi_{Rt}\psi^\dagger_{L0}\rangle \\ &- \langle \psi^\dagger_{Rt}\psi_{Lt}\rangle\langle \psi^\dagger_{R0}\psi_{L0}\rangle - \langle \psi^\dagger_{Rt}\psi_{L0}\rangle\langle \psi_{Lt}\psi^\dagger_{R0}\rangle \\ &+ \langle \psi^\dagger_{Lt}\psi_{Rt}\rangle\langle \psi^\dagger_{L0}\psi_{R0}\rangle + \langle \psi^\dagger_{Rt}\psi_{Lt}\rangle\langle \psi^\dagger_{R0}\psi_{L0}\rangle \\ &- \langle \psi^\dagger_{Lt}\psi_{Rt}\rangle\langle \psi^\dagger_{R0}\psi_{L0}\rangle - \langle \psi^\dagger_{Rt}\psi_{Lt}\rangle\langle \psi^\dagger_{L0}\psi_{R0}\rangle. \end{aligned} \tag{2.3.86}$$

The disconnected term on the first line cancels the first term on the last line; the disconnected term on the second line cancels the second term on the last line. The first terms on lines three and four cancel the first and second terms, respectively, on line five. Four connected contractions remain: the second terms on lines one through four, with signs $+, +, -, -$. The mixed connected contractions on lines three and four must be retained for the exact dressed result. Each mixed Green's function begins at order V_T, so those two products contribute to C_{II} from order V_T^4. The diagonal products already contribute at order V_T^2, but their dressed functions also contain higher orders. A leading-order approximation replaces the diagonal G's by disconnected g's and then drops the mixed products consistently.

2.3.7 Frequency integration and the dressed contact functions

Translate each surviving contraction using the index convention

$$\langle \psi^\dagger_i(t)\psi_j(0)\rangle = -iG^<_{ji}(0,t), \qquad \langle \psi_i(t)\psi^\dagger_j(0)\rangle = iG^>_{ij}(t,0). \tag{2.3.87}$$

The product $(-i)(i) = 1$ leaves the four Wick signs unchanged.

$$\begin{aligned}C_{II}(t,0) = e^2V_T^2\big[&G^<_{LL}(0,t)G^>_{RR}(t,0) + G^<_{RR}(0,t)G^>_{LL}(t,0) \\ &- G^<_{RL}(0,t)G^>_{RL}(t,0) - G^<_{LR}(0,t)G^>_{LR}(t,0)\big]. \end{aligned} \tag{2.3.88}$$

The diagonal products provide the leading contribution, while the mixed products first contribute at higher order. These labels describe the first power of V_T in each term; the displayed dressed expression keeps the complete functions.

$$\begin{aligned}C_{II}(t) = e^2V_T^2\big[&G^<_{LL}(-t)G^>_{RR}(t) + G^<_{RR}(-t)G^>_{LL}(t) \\ &- G^<_{RL}(-t)G^>_{RL}(t) - G^<_{LR}(-t)G^>_{LR}(t)\big]. \end{aligned} \tag{2.3.89}$$

The reduction to one time difference assumes a stationary state. It is not needed for (2.3.88). An independent compact check uses a matrix in the physical lead space:

$$\mathsf{J} = \begin{pmatrix} 0 & ieV_T \\ -ieV_T & 0 \end{pmatrix}, \qquad C_{II}(t,0) = \mathrm{Tr}_{L,R}\big[\mathsf{J}G^>(t,0)\mathsf{J}G^<(0,t)\big]. \tag{2.3.90}$$

Multiplying these two-by-two matrices reproduces precisely the four terms in (2.3.88); this trace is unrelated to contour space.

$$\mathcal{S}^{(2)}(0) = 2\int_{-\infty}^{\infty} dt\, C_{II}(t). \tag{2.3.91}$$

No factor $e^{i\Omega t}$ is present in this integral, so it evaluates the zero-frequency spectrum. The elementary product transform is

$$\begin{aligned}\int dt\, A(-t)B(t) &= \int dt \int \frac{d\omega_1 d\omega_2}{(2\pi)^2} A(\omega_1)B(\omega_2)e^{i(\omega_1-\omega_2)t} \\ &= \int \frac{d\omega_1 d\omega_2}{(2\pi)^2} A(\omega_1)B(\omega_2)2\pi\delta(\omega_1-\omega_2) \\ &= \frac{1}{2\pi}\int d\omega\, A(\omega)B(\omega).\end{aligned} \tag{2.3.92}$$

Combining this measure with the factor 2 in (2.3.91) gives

$$\begin{aligned}\mathcal{S}^{(2)}(0) = \frac{e^2V_T^2}{\pi}\int d\omega\, \big[&G^<_{LL}(\omega)G^>_{RR}(\omega) + G^<_{RR}(\omega)G^>_{LL}(\omega) \\ &- G^<_{RL}(\omega)G^>_{RL}(\omega) - G^<_{LR}(\omega)G^>_{LR}(\omega)\big].\end{aligned} \tag{2.3.93}$$

For comparison, at finite measurement frequency the ordered spectrum has a shifted internal argument:

$$\widetilde{C}_{II}(\Omega) = \int \frac{d\omega}{2\pi}\,\mathrm{Tr}_{L,R}\big[\mathsf{J}G^>(\omega+\Omega)\mathsf{J}G^<(\omega)\big]. \tag{2.3.94}$$

The half-symmetrized spectrum is $[\widetilde{C}_{II}(\Omega) + \widetilde{C}_{II}(-\Omega)]/2$. Consequently, doubling the ordered spectrum at general Ω does not give the doubled symmetrized spectrum. The subsequent calculation uses only $\Omega = 0$.

Substituting the full contact Green's functions. Set $D^R = 1 - V_T^2 g_L^R g_R^R$. The lesser/greater Dyson solution can be evaluated by an ordinary lead-space product:

$$\begin{aligned}G^{\lessgtr} &= \mathsf{B}^R \begin{pmatrix} g_L^{\lessgtr} & 0 \\ 0 & g_R^{\lessgtr}\end{pmatrix} \mathsf{B}^A, \qquad \mathsf{B}^A = (\mathsf{B}^R)^\dagger, \\ \mathsf{B}^R &= 1 + G^R\mathsf{V} = \frac{1}{D^R}\begin{pmatrix} 1 & V_T g_L^R \\ V_T g_R^R & 1\end{pmatrix}.\end{aligned} \tag{2.3.95}$$

For example, $1 + V_T G^R_{LR} = 1 + V_T^2 g_L^R g_R^R/D^R = 1/D^R$. The matrix B^R is an algebraic dressing factor, not a unitary scattering matrix. Its product with the populated reference functions is what carries the two distinct Fermi distributions into G. The six functions explicitly required by the noise calculation are

$$G^<_{LL} = \frac{g_L^< + V_T^2 g_L^R g_R^< g_L^A}{|1 - V_T^2 g_L^R g_R^R|^2}, \tag{2.3.96}$$

$$G^>_{LL} = \frac{g_L^> + V_T^2 g_L^R g_R^> g_L^A}{|1 - V_T^2 g_L^R g_R^R|^2}, \tag{2.3.97}$$

$$G^>_{RL} = \frac{V_T(g_R^R g_L^> + g_R^> g_L^A)}{|1 - V_T^2 g_L^R g_R^R|^2}, \tag{2.3.98}$$

$$G^>_{LR} = \frac{V_T(g_L^R g_R^> + g_L^> g_R^A)}{|1 - V_T^2 g_L^R g_R^R|^2}, \tag{2.3.99}$$

$$G^<_{RR} = \frac{g_R^< + V_T^2 g_R^R g_L^< g_R^A}{|1 - V_T^2 g_L^R g_R^R|^2}, \tag{2.3.100}$$

$$G^>_{RR} = \frac{g_R^> + V_T^2 g_R^R g_L^> g_R^A}{|1 - V_T^2 g_L^R g_R^R|^2}. \tag{2.3.101}$$

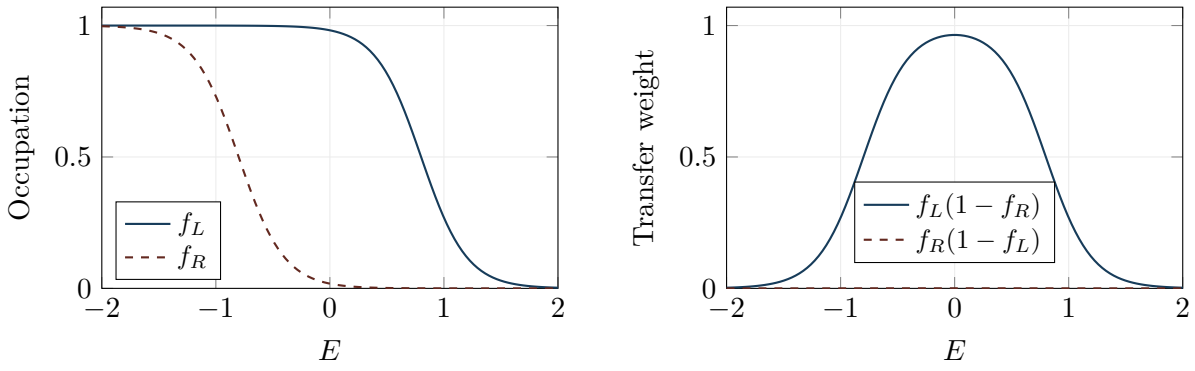


Figure 2.19: Fermionic occupations and transfer windows, evaluated from $f_\ell(E) = [e^{(E-\mu_\ell)/T} + 1]^{-1}$ with $\mu_L = 0.8$, $\mu_R = -0.8$, and $T = 0.2$ in the same energy units. The right panel shows the occupied-to-empty weights $f_L(1 - f_R)$ and $f_R(1 - f_L)$. Their difference drives the mean current; their sum counts both directions of fluctuation. Pauli exclusion suppresses transfers when both states are occupied or both are empty. These are analytic model curves.

The two mixed lesser entries follow from exactly the same multiplication:

$$G^<_{RL} = \frac{V_T(g^R_R g^<_L + g^<_R g^A_L)}{|D^R|^2}, \qquad G^<_{LR} = \frac{V_T(g^R_L g^<_R + g^<_L g^A_R)}{|D^R|^2}. \tag{2.3.102}$$

For a complex hopping, V_T in the RL numerator becomes V_T^*; all denominator and diagonal occurrences become $|V_T|^2$. These changes follow from Hermiticity of the contact Hamiltonian. Use the equilibrium relations again for each disconnected lead:

$$g^<_\ell(\omega) = f_\ell(\omega)[g^A_\ell(\omega) - g^R_\ell(\omega)], \tag{2.3.103}$$

$$g^>_\ell(\omega) = -[1 - f_\ell(\omega)][g^A_\ell(\omega) - g^R_\ell(\omega)]. \tag{2.3.104}$$

These relations apply separately to g_L and g_R. A single scalar Fermi function cannot be assigned to the full biased contact. Each transfer probability contains an occupied initial state and an empty final state, $f_L(1 - f_R)$ or $f_R(1 - f_L)$. Their difference is $f_L - f_R$, while their sum enters the noise.

2.3.8 The full polynomial reduction to Landauer noise

Continue with identical wide-band leads and real V_T. Every function in the following algebra is evaluated at the same internal frequency ω; write $f_\ell = f_\ell(\omega)$. The denominator is $|D^R|^2 = (1 + \lambda_c)^2$. For example,

$$V_T^2 g^R_L g^<_R g^A_L = V_T^2(-i\pi\nu_0)(2\pi i\nu_0 f_R)(i\pi\nu_0) = 2\pi i\nu_0\lambda_c f_R. \tag{2.3.105}$$

$$G^<_{LL} = \frac{2\pi i\nu_0}{(1+\lambda_c)^2}(f_L + \lambda_c f_R). \tag{2.3.106}$$

$$G^>_{RR} = \frac{2\pi i\nu_0}{(1+\lambda_c)^2}[(f_R - 1) + \lambda_c(f_L - 1)]. \tag{2.3.107}$$

Passing from the first line to the second interchanges the reservoir labels and replaces each occupation by $f_\ell - 1$. The other two diagonal functions follow by the same interchange.

Diagonal products and their full expansion.

$$\begin{aligned} &G^<_{LL}G^>_{RR} + G^<_{RR}G^>_{LL} \\ &\quad = -\frac{4\pi^2\nu_0^2}{(1+\lambda_c)^4}\Big\{(f_L + \lambda_c f_R)[f_R - 1 + \lambda_c(f_L - 1)] \\ &\qquad\qquad + (f_R + \lambda_c f_L)[f_L - 1 + \lambda_c(f_R - 1)]\Big\}. \end{aligned} \tag{2.3.108}$$

To see the powers of contact strength individually, distribute each product:

$$(f_L + \lambda_c f_R)[f_R - 1 + \lambda_c(f_L - 1)] = f_L(f_R - 1) + \lambda_c f_L(f_L - 1) + \lambda_c f_R(f_R - 1) + \lambda_c^2 f_R(f_L - 1), \tag{2.3.109}$$

$$(f_R + \lambda_c f_L)[f_L - 1 + \lambda_c(f_R - 1)] = f_R(f_L - 1) + \lambda_c f_R(f_R - 1) + \lambda_c f_L(f_L - 1) + \lambda_c^2 f_L(f_R - 1). \tag{2.3.110}$$

Collecting the constant, quadratic, and linear powers in that order gives

$$G^<_{LL}G^>_{RR} + G^<_{RR}G^>_{LL} = -\frac{4\pi^2\nu_0^2}{(1+\lambda_c)^4}\Big\{ f_L(f_R - 1) + f_R(f_L - 1) + \lambda_c^2[f_R(f_L - 1) + f_L(f_R - 1)] + \lambda_c[f_R^2 - f_R + f_L^2 - f_L + f_R^2 - f_R + f_L^2 - f_L]\Big\}. \tag{2.3.111}$$

The overall minus sign converts $f(f' - 1)$ to $f(1 - f')$:

$$G^<_{LL}G^>_{RR} + G^<_{RR}G^>_{LL} = \frac{4\pi^2\nu_0^2}{(1+\lambda_c)^4}\Big\{[f_L(1 - f_R) + f_R(1 - f_L)](1 + \lambda_c^2) + 2\lambda_c(-f_L^2 - f_R^2 + f_L + f_R)\Big\}. \tag{2.3.112}$$

Mixed products and the square of the bias window.

$$G^<_{RL} = \frac{2\pi^2\nu_0^2 V_T}{(1+\lambda_c)^2}(f_L - f_R). \tag{2.3.113}$$

$$G^>_{RL} = \frac{2\pi^2\nu_0^2 V_T}{(1+\lambda_c)^2}(f_L - f_R). \tag{2.3.114}$$

For this wide-band contact, $G^R_{RL} = G^A_{RL}$, so $G^>_{RL} - G^<_{RL} = 0$. The adjoint identities give $G^{\lessgtr}_{LR} = -G^{\lessgtr}_{RL}$, since these mixed lesser/greater entries are real for the chosen hopping phase. Consequently,

$$-[G^<_{RL}G^>_{RL} + G^<_{LR}G^>_{LR}] = -2\left[\frac{2\pi^2\nu_0^2 V_T}{(1+\lambda_c)^2}\right]^2 (f_L - f_R)^2 = -\frac{8\pi^2\nu_0^2\lambda_c}{(1+\lambda_c)^4}(f_L^2 + f_R^2 - 2f_L f_R). \tag{2.3.115}$$

The same coefficient can be displayed as the difference used in the polynomial rearrangement:

$$-[G^<_{RL}G^>_{RL} + G^<_{LR}G^>_{LR}] = \frac{4\pi^2\nu_0^2}{(1+\lambda_c)^4}(f_L^2 + f_R^2 - 2f_L f_R)(2\lambda_c - 4\lambda_c). \tag{2.3.116}$$

The relation between the squared occupations and the diagonal polynomial is made explicit by introducing

$$X = f_L(1 - f_R) + f_R(1 - f_L), \quad Y = f_L(1 - f_L) + f_R(1 - f_R), \quad Q = (f_L - f_R)^2 = f_L^2 + f_R^2 - 2f_L f_R, \qquad X = Y + Q. \tag{2.3.117}$$

These symbols only group terms; they do not change the approximation. In particular,

$$(1 + \lambda_c^2)X + 2\lambda_c Y = (1 + \lambda_c^2)X + 2\lambda_c(X - Q) = (1 + \lambda_c)^2 X - 2\lambda_c Q. \tag{2.3.118}$$

The mixed products provide another $-2\lambda_c Q$, yielding $(1+\lambda_c)^2 X - 4\lambda_c Q$ in the common numerator.

$$\begin{aligned} &G^<_{LL}G^>_{RR} + G^<_{RR}G^>_{LL} - G^<_{RL}G^>_{RL} - G^<_{LR}G^>_{LR} \\ &\quad = \frac{4\pi^2\nu_0^2}{(1+\lambda_c)^2}[f_L(1-f_R) + f_R(1-f_L)] - \frac{16\pi^2\nu_0^2\lambda_c}{(1+\lambda_c)^4}(f_L - f_R)^2. \end{aligned} \tag{2.3.119}$$

Transmission form and physical limits. Substitution into (2.3.93) gives two integrals:

$$\begin{aligned} \mathcal{S}^{(2)}(0) = \frac{e^2}{\pi}\frac{4\lambda_c}{(1+\lambda_c)^2}\int d\omega\,[f_L(1-f_R) + f_R(1-f_L)] \\ - \frac{e^2}{\pi}\left[\frac{4\lambda_c}{(1+\lambda_c)^2}\right]^2 \int d\omega\,(f_L - f_R)^2. \end{aligned} \tag{2.3.120}$$

The first coefficient contains $\mathcal{T}$, and the second contains $\mathcal{T}^2$. This is the Landauer description of current noise. With units restored and the integration variable written as energy E, the two equivalent forms are

$$\mathcal{S}^{(2)}(0) = \frac{2e^2}{h}\int dE\,\{\mathcal{T}[f_L(1-f_R) + f_R(1-f_L)] - \mathcal{T}^2(f_L - f_R)^2\}, \tag{2.3.121}$$

$$= \frac{2e^2}{h}\int dE\,\{\mathcal{T}[f_L(1-f_L) + f_R(1-f_R)] + \mathcal{T}(1-\mathcal{T})(f_L - f_R)^2\}. \tag{2.3.122}$$

The second follows directly from $X = Y + Q$. It separates thermal occupation fluctuations from partition fluctuations and makes nonnegativity manifest for $0 \le \mathcal{T} \le 1$. The constant transmission of the present contact can be taken outside the integrals. In a general normal elastic single-channel scatterer, the same Landauer form uses $\mathcal{T}(E)$ inside the integral; its microscopic energy dependence must be computed for that scatterer.

At equilibrium, $f_L = f_R = f$, so only the thermal term remains. Using

$$f(1-f) = -k_B T\partial_E f, \qquad \int dE\,f(1-f) = k_B T, \tag{2.3.123}$$

gives, for the matching linear conductance,

$$\mathcal{S}^{(2)}(0) = 4k_B T\mathcal{G}, \qquad \mathcal{S}^{\mathrm{sym}}(0) = 2k_B T\mathcal{G}. \tag{2.3.124}$$

This equilibrium identity also holds for energy-dependent transmission, with the corresponding conductance

$$\mathcal{G} = \frac{e^2}{h}\int dE\,\mathcal{T}(E)(-\partial_E f). \tag{2.3.125}$$

At zero temperature, $f_\ell(1-f_\ell) = 0$ except on a set of zero measure, and $(f_L - f_R)^2$ is one throughout the bias window of width $|\Delta\mu|$. For constant transmission,

$$\mathcal{S}^{(2)}(0) = \frac{2e^2}{h}\mathcal{T}(1-\mathcal{T})|\Delta\mu| = 2e|I|(1-\mathcal{T}), \qquad \mathcal{S}^{\mathrm{sym}}(0) = e|I|(1-\mathcal{T}). \tag{2.3.126}$$

The Fano factor in the doubled convention is $F = \mathcal{S}^{(2)}(0)/(2e|I|) = 1 - \mathcal{T}$, for nonzero current. Rare tunnelling events approach the Poisson value $F = 1$. Perfect transmission removes zero-temperature partition noise, but thermal noise survives at nonzero temperature. Reservoir interchange reverses I while leaving $\mathcal{S}$ unchanged.

The continuous thermal-to-shot crossover can be evaluated without numerical integration. For common $T > 0$, put $\Delta = \Delta\mu$. Direct Fermi algebra gives

$$f_L(1-f_R) = \frac{f_L - f_R}{1 - e^{-\beta\Delta}}, \qquad f_R(1-f_L) = \frac{f_L - f_R}{e^{\beta\Delta} - 1}. \tag{2.3.127}$$

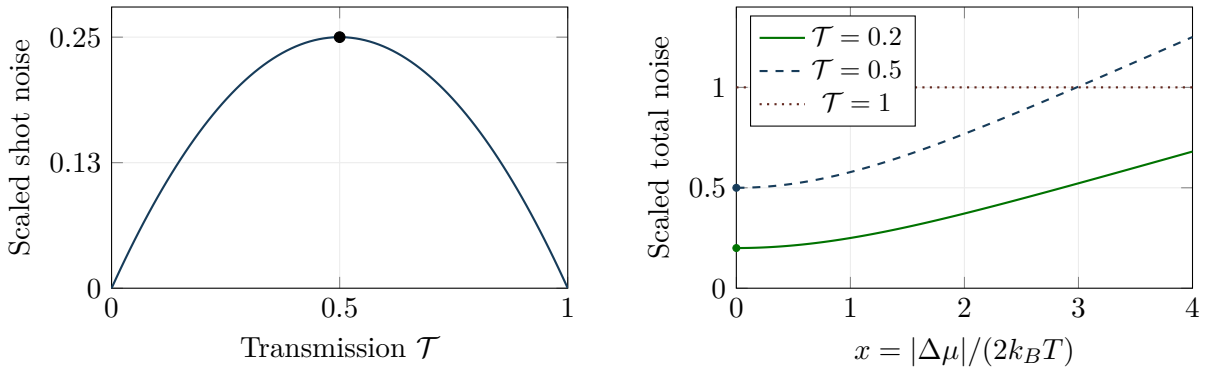


Figure 2.20: Analytic noise checks. Left: at $T = 0$ and fixed nonzero bias, $h\mathcal{S}^{(2)}/(2e^2|\Delta\mu|) = \mathcal{T}(1-\mathcal{T})$, with maximum at $\mathcal{T} = 1/2$. Right: the finite-temperature crossover $\mathcal{S}^{(2)}/[4k_BT(e^2/h)] = \mathcal{T}^2 + \mathcal{T}(1-\mathcal{T})x\coth x$, where $x = |\Delta\mu|/(2k_BT)$, for transmissions $0.2, 0.5, 1$. The $x = 0$ value is the equilibrium conductance in units e^2/h. The unit-transmission curve stays thermal and independent of bias in this constant-transmission model.

Their sum integrates to $\Delta\coth[\Delta/(2k_BT)]$, while $\int dE\, Y = 2k_BT$. Therefore

$$\mathcal{S}^{(2)}(0) = \frac{2e^2}{h}\left[2k_BT\mathcal{T}^2 + \mathcal{T}(1-\mathcal{T})\Delta\coth\left(\frac{\Delta}{2k_BT}\right)\right]. \tag{2.3.128}$$

At $\Delta = 0$, use the continuous limit $\Delta\coth[\Delta/(2k_BT)] \to 2k_BT$.

Finally, a comparison with a tunnelling expansion must use the same order. Since $\mathcal{T} = 4\lambda_c + O(\lambda_c^2)$, define $\mathcal{T}_0 = 4\pi^2|V_T|^2\nu_L\nu_R$. Then

$$\mathcal{S}^{(2)}(0) = \frac{2e^2}{h}\mathcal{T}_0\int dE\,[f_L(1-f_R) + f_R(1-f_L)] + O(|V_T|^4). \tag{2.3.129}$$

The gauge-transformed calculation that follows must reproduce this weak-contact limit. The full $\mathcal{T}^2$ subtraction and all-order denominator cannot be inferred from a leading-order calculation alone.

2.4 Bias as a tunnelling phase

The last calculation represents the same biased contact in a different energy convention. In the physical-energy description the two reservoirs have distributions $f_\ell(E) = f_0(E-\mu_\ell)$ and the hopping is constant. In a shifted-energy description, the free occupations both take the function $f_0(\xi) = (e^{\beta\xi}+1)^{-1}$, but the hopping acquires a relative phase. This is a change of representation. The reservoirs retain their distinct chemical potentials and do not become a globally equilibrated device.

Let $H_0 = H_L + H_R$, $M = \mu_L N_L + \mu_R N_R$, and $\mathcal{K}_0 = H_0 - M$. The free reservoirs conserve their particle numbers, so $[H_0, M] = 0$. Put $s = t - t_0$. The commutator $[N_\ell, \psi_\ell] = -\psi_\ell$ gives

$$e^{-iMs}\psi_\ell e^{iMs} = \sum_{n=0}^{\infty}\frac{(-is)^n}{n!}\underbrace{[M,[M,\ldots[M,\psi_\ell]]]}_{n\text{ commutators}} = e^{+i\mu_\ell s}\psi_\ell, \tag{2.4.1}$$

$$\widetilde{\psi}_\ell(t) \equiv e^{i\mathcal{K}_0 s}\psi_\ell e^{-i\mathcal{K}_0 s} = e^{iH_0 s}e^{-iMs}\psi_\ell e^{iMs}e^{-iH_0 s} = e^{+i\mu_\ell s}\psi_{\ell,H_0}(t), \tag{2.4.2}$$

$$\psi_{\ell,H_0}(t) = e^{-i\mu_\ell s}\widetilde{\psi}_\ell(t), \qquad \psi^\dagger_{\ell,H_0}(t) = e^{+i\mu_\ell s}\widetilde{\psi}^\dagger_\ell(t). \tag{2.4.3}$$

The $n = 0$ term in the commutator expansion denotes ψ_ℓ itself. Both the annihilator and creator phases are therefore fixed before any noise integral is evaluated. The coordinates of an energy level change as $\xi_{\ell k} = \epsilon_{\ell k} - \mu_\ell$. For a free stationary two-point function the two field phases combine into $e^{i\mu_\ell(t-t')}$, and hence

$$\widetilde{g}^{\lessgtr}_\ell(\tau) = e^{i\mu_\ell\tau}g^{\lessgtr}_{\ell,H_0}(\tau), \qquad \widetilde{g}^{\lessgtr}_\ell(\omega) = g^{\lessgtr}_{\ell,H_0}(\omega+\mu_\ell). \tag{2.4.4}$$

2.4.1 The tunnelling Hamiltonian and current

At the contact $x = 0$, the tunnelling Hamiltonian is

$$H_T = V_T \psi_L^\dagger \psi_R + V_T^* \psi_R^\dagger \psi_L. \tag{2.4.5}$$

We keep a general complex V_T here; the real-amplitude phase choice used in the preceding polynomial calculation is a special case. Substitution of each of the two field transformations gives

$$V_T \psi_{L,H_0}^\dagger \psi_{R,H_0} = V_T e^{i\mu_L s} e^{-i\mu_R s} \widetilde{\psi}_L^\dagger \widetilde{\psi}_R, \qquad V_T^* \psi_{R,H_0}^\dagger \psi_{L,H_0} = V_T^* e^{i\mu_R s} e^{-i\mu_L s} \widetilde{\psi}_R^\dagger \widetilde{\psi}_L. \tag{2.4.6}$$

With $\Delta\mu = \mu_L - \mu_R$, the two terms read

$$\begin{aligned} \widetilde{H}_T(t) = {} & V_T e^{+i\Delta\mu s} \widetilde{\psi}_L^\dagger(t) \widetilde{\psi}_R(t) \\ & + V_T^* e^{-i\Delta\mu s} \widetilde{\psi}_R^\dagger(t) \widetilde{\psi}_L(t). \end{aligned} \tag{2.4.7}$$

From here on all fields in this calculation are understood in the shifted representation. We keep the tilde to make that convention explicit. The tunnelling current follows from the same number commutator as before.

$$\widetilde{I}(t) = ie \left[V_T e^{+i\Delta\mu s} \widetilde{\psi}_L^\dagger(t) \widetilde{\psi}_R(t) - V_T^* e^{-i\Delta\mu s} \widetilde{\psi}_R^\dagger(t) \widetilde{\psi}_L(t) \right]. \tag{2.4.8}$$

The $\widetilde{\psi}_L^\dagger \widetilde{\psi}_R$ operator transfers an electron from R to L; its conjugate transfers one from L to R. The current convention $I = -e\dot{N}_L$ is unchanged. The phases describe relative energy reference, rather than an extra driving voltage added to the physical bias already present in the occupations.

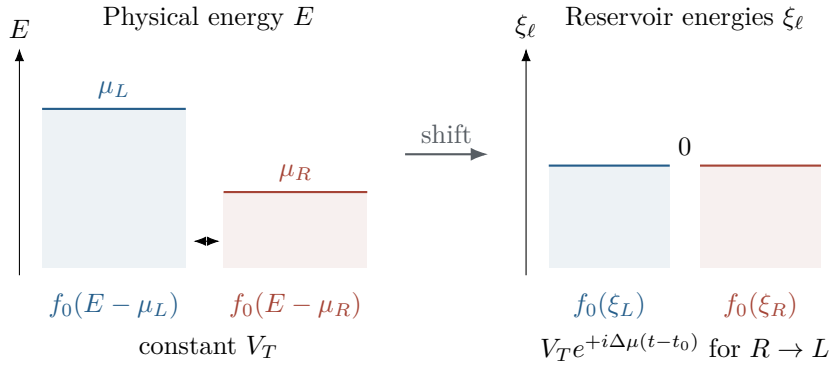


Figure 2.21: Two equivalent descriptions of the same contact (schematic). On the left, physical energies share one axis and the occupations have different chemical potentials. On the right, each reservoir uses $\xi_\ell = E - \mu_\ell$ and the common function $f_0(\xi_\ell)$. The phase multiplying $\widetilde{\psi}_L^\dagger \widetilde{\psi}_R$ is $e^{+i\Delta\mu(t-t_0)}$; the reverse term has its complex conjugate. The energy-dependent density of states must shift along with its energy argument.

2.4.2 Ordered noise and the leading tunnelling order

To expose the disconnected term, define $X(t) = \widetilde{\psi}_L^\dagger(t) \widetilde{\psi}_R(t)$, $Y(t) = X^\dagger(t)$, $a(t) = V_T e^{i\Delta\mu(t-t_0)}$, and $B(t) = a(t)X(t) - a^*(t)Y(t)$, so that $\widetilde{I} = ieB$. At finite coupling, the brackets include the interaction-picture evolution generated by $\widetilde{H}_T$. At the leading order retained below they reduce to the disconnected reference average. Direct multiplication of the two currents gives the

cross-transfer terms below, a disconnected subtraction, and two same-direction terms $\mathcal{R}_{\text{same}}$:

$$\begin{aligned} C_{II}(t,t') &= e^2|V_T|^2 e^{+i\Delta\mu(t-t')}\langle\widetilde{\psi}_L^\dagger(t)\widetilde{\psi}_R(t)\widetilde{\psi}_R^\dagger(t')\widetilde{\psi}_L(t')\rangle \\ &\quad + e^2|V_T|^2 e^{-i\Delta\mu(t-t')}\langle\widetilde{\psi}_R^\dagger(t)\widetilde{\psi}_L(t)\widetilde{\psi}_L^\dagger(t')\widetilde{\psi}_R(t')\rangle \\ &\quad + e^2\Big[V_T e^{+i\Delta\mu(t-t_0)}\langle\widetilde{\psi}_L^\dagger(t)\widetilde{\psi}_R(t)\rangle \\ &\qquad\qquad - V_T^* e^{-i\Delta\mu(t-t_0)}\langle\widetilde{\psi}_R^\dagger(t)\widetilde{\psi}_L(t)\rangle\Big] \\ &\quad \times\Big[V_T e^{+i\Delta\mu(t'-t_0)}\langle\widetilde{\psi}_L^\dagger(t')\widetilde{\psi}_R(t')\rangle \\ &\qquad\qquad - V_T^* e^{-i\Delta\mu(t'-t_0)}\langle\widetilde{\psi}_R^\dagger(t')\widetilde{\psi}_L(t')\rangle\Big] + \mathcal{R}_{\text{same}}(t,t'). \end{aligned} \tag{2.4.9}$$

Here the operator products in the first two lines are explicitly

$$X(t)Y(t') = \widetilde{\psi}_L^\dagger(t)\widetilde{\psi}_R(t)\widetilde{\psi}_R^\dagger(t')\widetilde{\psi}_L(t'), \tag{2.4.10}$$

$$Y(t)X(t') = \widetilde{\psi}_R^\dagger(t)\widetilde{\psi}_L(t)\widetilde{\psi}_L^\dagger(t')\widetilde{\psi}_R(t'), \tag{2.4.11}$$

$$\mathcal{R}_{\text{same}}(t,t') = -e^2 a(t)a(t')\langle X(t)X(t')\rangle - e^2 a^*(t)a^*(t')\langle Y(t)Y(t')\rangle. \tag{2.4.12}$$

The plus sign preceding the product of averages in (2.4.9) is required: $-\langle I(t)\rangle\langle I(t')\rangle = -i^2e^2\langle B(t)\rangle\langle B(t')\rangle$.

The product of averages contributes only at higher order. We now focus on the leading order in $|V_T|^2$. The reference state is the product of two normal Gaussian equilibrium reservoirs. Separate particle-number conservation gives $\langle X\rangle_0 = \langle Y\rangle_0 = 0$ and $\langle XX\rangle_0 = \langle YY\rangle_0 = 0$. In the coupled expansion $\langle X\rangle = O(V_T^*)$ and $\langle Y\rangle = O(V_T)$, so the mean current is $O(|V_T|^2)$ and its product is $O(|V_T|^4)$. Nonzero same-direction four-point averages need two additional tunnelling insertions; their contribution to noise also starts at $O(|V_T|^4)$. These are order statements for normal, number-conserving leads and this perturbative preparation.

The noise in the time domain remains the ordered connected function

$$C_{II}(t,t') = \langle\delta I(t)\delta I(t')\rangle. \tag{2.4.13}$$

Wick factorization of the cross-transfer averages is particularly transparent: moving $\widetilde{\psi}_L(t')$ across the two R operators costs $(-1)^2 = +1$, after which the factorized reference state separates the leads. The other transfer direction works the same way. Thus

$$\begin{aligned} C_{II}(t,t') = e^2|V_T|^2\Big[&e^{+i\Delta\mu(t-t')}\langle\widetilde{\psi}_L^\dagger(t)\widetilde{\psi}_L(t')\rangle_0\langle\widetilde{\psi}_R(t)\widetilde{\psi}_R^\dagger(t')\rangle_0 \\ &+ e^{-i\Delta\mu(t-t')}\langle\widetilde{\psi}_L(t)\widetilde{\psi}_L^\dagger(t')\rangle_0\langle\widetilde{\psi}_R^\dagger(t)\widetilde{\psi}_R(t')\rangle_0\Big] + O(|V_T|^4). \end{aligned} \tag{2.4.14}$$

In the first product the occupied state is supplied *from* L and the empty state accepts it *into* R. In the second product the empty state is *in* L, while the occupied state comes *from* R. There is no contradiction with the operator order in the earlier four-point function: an ordered autocorrelation contains a transfer and its conjugate, and its occupation factor identifies the available initial state. Substitution of the definitions gives

$$\langle\widetilde{\psi}_L^\dagger(t)\widetilde{\psi}_L(t')\rangle_0 = -i\widetilde{g}_L^<(t',t), \qquad \langle\widetilde{\psi}_R(t)\widetilde{\psi}_R^\dagger(t')\rangle_0 = +i\widetilde{g}_R^>(t,t'), \tag{2.4.15}$$

$$\langle\widetilde{\psi}_L(t)\widetilde{\psi}_L^\dagger(t')\rangle_0 = +i\widetilde{g}_L^>(t,t'), \qquad \langle\widetilde{\psi}_R^\dagger(t)\widetilde{\psi}_R(t')\rangle_0 = -i\widetilde{g}_R^<(t',t). \tag{2.4.16}$$

Both products of prefactors equal $+1$. Since the free reference leads are stationary, their arguments reduce to time differences:

$$\begin{aligned} C_{II}(\tau) = e^2|V_T|^2\Big[&e^{+i\Delta\mu\tau}\widetilde{g}_L^<(-\tau)\widetilde{g}_R^>(\tau) \\ &+ e^{-i\Delta\mu\tau}\widetilde{g}_L^>(\tau)\widetilde{g}_R^<(-\tau)\Big] + O(|V_T|^4), \qquad \tau = t-t'. \end{aligned} \tag{2.4.17}$$

2.4.3 Fourier integrals, delta functions, and energy shifts

The doubled zero-frequency noise is

$$\mathcal{S}^{(2)}(0) = 2\int_{-\infty}^{\infty} d(t-t')\, C_{II}(t,t'). \tag{2.4.18}$$

This equation uses stationarity and $\Omega = 0$. An ordered spectrum at nonzero measurement frequency contains $e^{i\Omega(t-t')}$; its symmetrization combines $+\Omega$ and $-\Omega$. The factor of two in (2.4.18) therefore cannot be carried over to a general ordered finite-frequency spectrum without symmetrizing it.

Inserting the time-domain correlator, with both transfer directions retained, gives at leading order

$$\begin{aligned}\mathcal{S}^{(2)}(0) = 2e^2|V_T|^2 \int d(t-t')\Big[&e^{+i\Delta\mu(t-t')}\widetilde{g}_L^{<}(t'-t)\widetilde{g}_R^{>}(t-t') \\ &+ e^{-i\Delta\mu(t-t')}\widetilde{g}_L^{>}(t-t')\widetilde{g}_R^{<}(t'-t)\Big] + O(|V_T|^4).\end{aligned} \tag{2.4.19}$$

To keep the equalities below uncluttered, write $\mathcal{S}^{(2)} = \mathcal{S}^{(2)}_{\rm tun} + O(|V_T|^4)$. The subscript denotes the coefficient retained to order $|V_T|^2$. Rename the integration variable $\tau = t - t'$, with $d\tau = d(t-t')$:

$$\mathcal{S}^{(2)}_{\rm tun}(0) = 2e^2|V_T|^2 \int d\tau\, \left[e^{+i\Delta\mu\tau}\widetilde{g}_L^{<}(-\tau)\widetilde{g}_R^{>}(\tau) + e^{-i\Delta\mu\tau}\widetilde{g}_L^{>}(\tau)\widetilde{g}_R^{<}(-\tau)\right]. \tag{2.4.20}$$

The inverse Fourier transform is $g(\tau) = \int d\omega\, e^{-i\omega\tau} g(\omega)/(2\pi)$. Thus the negative time argument carries a positive Fourier phase. Keeping all three exponentials in each term makes the energy constraint visible:

$$\begin{aligned}\mathcal{S}^{(2)}_{\rm tun}(0) = \frac{2e^2|V_T|^2}{(2\pi)^2} \int d\tau\, d\omega_1\, d\omega_2 \Big[&e^{+i\Delta\mu\tau}e^{+i\omega_1\tau}e^{-i\omega_2\tau}\widetilde{g}_L^{<}(\omega_1)\widetilde{g}_R^{>}(\omega_2) \\ &+ e^{-i\Delta\mu\tau}e^{-i\omega_1\tau}e^{+i\omega_2\tau}\widetilde{g}_L^{>}(\omega_1)\widetilde{g}_R^{<}(\omega_2)\Big].\end{aligned} \tag{2.4.21}$$

For the first transfer direction the time integral is

$$\int d\tau\, e^{i(\Delta\mu+\omega_1-\omega_2)\tau} = 2\pi\delta(\Delta\mu+\omega_1-\omega_2). \tag{2.4.22}$$

For the second it is

$$\int d\tau\, e^{-i(\Delta\mu+\omega_1-\omega_2)\tau} = 2\pi\delta(-\Delta\mu-\omega_1+\omega_2) = 2\pi\delta(\Delta\mu+\omega_1-\omega_2), \tag{2.4.23}$$

because the delta function is even. Consequently

$$\begin{aligned}\mathcal{S}^{(2)}_{\rm tun}(0) = \frac{2e^2|V_T|^2}{2\pi} \int d\omega_1 d\omega_2 \Big[&\delta(\omega_1-\omega_2+\mu_L-\mu_R)\widetilde{g}_L^{<}(\omega_1)\widetilde{g}_R^{>}(\omega_2) \\ &+ \delta(\omega_1-\omega_2+\mu_L-\mu_R)\widetilde{g}_L^{>}(\omega_1)\widetilde{g}_R^{<}(\omega_2)\Big].\end{aligned} \tag{2.4.24}$$

The root of either constraint is $\omega_2 = \omega_1 + \mu_L - \mu_R$. The derivative of its argument with respect to ω_2 is -1, so $\delta(F(\omega_2)) = \delta(\omega_2 - \omega_1 - \Delta\mu)/|-1|$. No additional Jacobian is generated by the ω_2 integral.

$$\begin{aligned}\mathcal{S}^{(2)}_{\rm tun}(0) = \frac{2e^2|V_T|^2}{2\pi} \int d\omega\, \Big[&\widetilde{g}_L^{<}(\omega)\widetilde{g}_R^{>}(\omega+\Delta\mu) \\ &+ \widetilde{g}_L^{>}(\omega)\widetilde{g}_R^{<}(\omega+\Delta\mu)\Big].\end{aligned} \tag{2.4.25}$$

Here the remaining dummy variable ω_1 has been renamed ω. The two shifted energies differ by $\Delta\mu$, while their physical energies coincide: $\omega + \mu_L = (\omega + \Delta\mu) + \mu_R$. Set $E = \omega + \mu_L$, so that $dE = d\omega$. The next change of variables is therefore

$$\mathcal{S}^{(2)}_{\rm tun}(0) = \frac{2e^2|V_T|^2}{2\pi}\int dE\,\Big[\widetilde{g}^{<}_L(E-\mu_L)\widetilde{g}^{>}_R(E-\mu_R) + \widetilde{g}^{>}_L(E-\mu_L)\widetilde{g}^{<}_R(E-\mu_R)\Big]. \tag{2.4.26}$$

For free leads with local density of states $\nu_\ell(E)$,

$$\widetilde{g}^{<}_\ell(E-\mu_\ell) = 2\pi i\nu_\ell(E) f_0(E-\mu_\ell), \tag{2.4.27}$$

$$\widetilde{g}^{>}_\ell(E-\mu_\ell) = -2\pi i\nu_\ell(E)[1-f_0(E-\mu_\ell)]. \tag{2.4.28}$$

In particular, $\widetilde{\nu}_\ell(\xi) = \nu_\ell(\xi+\mu_\ell)$: the density of states changes its argument together with the occupations. Multiplication gives $(2\pi i)(-2\pi i) = 4\pi^2$ in each term.

$$\mathcal{S}^{(2)}_{\rm tun}(0) = \frac{2e^2}{2\pi}\int dE\,4\pi^2|V_T|^2\nu_L(E)\nu_R(E)\Big\{f_0(E-\mu_L)[1-f_0(E-\mu_R)] + f_0(E-\mu_R)[1-f_0(E-\mu_L)]\Big\}. \tag{2.4.29}$$

For identical flat leads the combination in front of the occupations is $4\pi^2|V_T|^2\nu_0^2 = \mathcal{T}_0 = 4\lambda_c$; the remaining noise coefficient is $2e^2/(2\pi) = e^2/\pi$ in $\hbar = 1$ units. The first product is $f_L(E)[1-f_R(E)]$: an occupied left state and an empty right state. Equivalently, the occupation integral can be written with the two transfer directions in the reverse order,

$$\int dE\,\Big\{f_0(E-\mu_R)[1-f_0(E-\mu_L)] + f_0(E-\mu_L)[1-f_0(E-\mu_R)]\Big\}. \tag{2.4.30}$$

The two occupied factors are respectively $f_R(E)$ and $f_L(E)$. The integration domain is the overlap of available lead states if their bands are finite. In the ideal flat wide-band limit it is the full real axis. Restoring $h = 2\pi\hbar$ yields

$$\boxed{\mathcal{S}^{\rm sym}_{\rm tun}(0) = \frac{e^2}{h}\int dE\,\mathcal{T}_0(E)\{f_L(1-f_R) + f_R(1-f_L)\}.} \tag{2.4.31}$$

Here $\mathcal{T}_0(E) = 4\pi^2|V_T|^2\nu_L(E)\nu_R(E)$. The doubled spectrum is twice this result. Every occupation and spectral factor in this expression uses the same physical energy E.

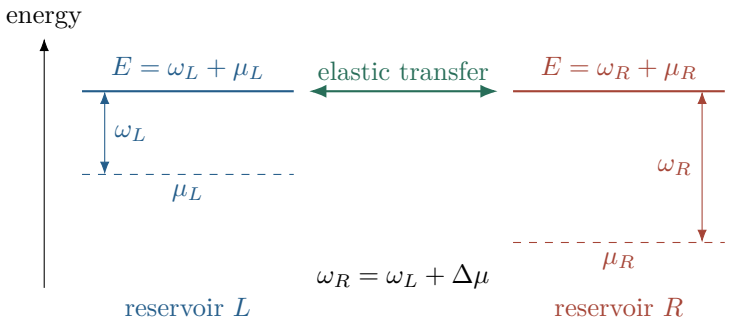


Figure 2.22: Energy conservation in the shifted representation (schematic). The Fourier delta function imposes $\omega_R = \omega_L + \Delta\mu$. Adding the respective chemical potentials puts both endpoints at the same physical energy E. The two products of Fermi functions count opposite transfer directions at this common energy; the relation remains valid with energy-dependent densities of states.

2.4.4 Comparison with the dressed noise calculation

The first derivation used the full coupled Gaussian functions and retained the contact denominator. The present one evaluates the contractions in the disconnected state. For the same flat, one-channel model,

$$\mathcal{T} = \frac{4\lambda_c}{(1+\lambda_c)^2} = 4\lambda_c - 8\lambda_c^2 + O(\lambda_c^3) = \mathcal{T}_0 + O(|V_T|^4), \tag{2.4.32}$$

$$\mathcal{T}^2 = O(|V_T|^4). \tag{2.4.33}$$

Insert these two expansions into the dressed expression, without changing its noise convention:

$$\mathcal{S}^{(2)}(0) = \frac{2e^2}{h}\int dE\ \left\{\mathcal{T}[f_L(1-f_R) + f_R(1-f_L)] - \mathcal{T}^2(f_L - f_R)^2\right\} \tag{2.4.34}$$

$$= \frac{2e^2}{h}\int dE\,\mathcal{T}_0[f_L(1-f_R) + f_R(1-f_L)] + O(|V_T|^4). \tag{2.4.35}$$

This is precisely twice (2.4.31) at the retained order. The negative partition correction and the denominator corrections start at $|V_T|^4$, so a lowest-order calculation cannot reproduce the perfect-transmission suppression on its own.

Object	Dressed contact calculation	Gauge calculation here
Reference information	Full coupled $G^{\lessgtr}$, built from populated $g^{\lessgtr}$	Disconnected $\widetilde{g}^{\lessgtr}$ at the retained order
Bias	$f_L(E)$ and $f_R(E)$	Shifted occupations plus conjugate hopping phases
Accuracy	Model-exact Gaussian contact resummation	Leading $\|V_T\|^2$ term
Noise dictionary	$\mathcal{S}^{(2)} = 2\mathcal{S}^{\mathrm{sym}}$	Same dictionary
Common limit	Expand $\mathcal{T}$ and $\mathcal{T}^2$ consistently	Equation (2.4.31)

Table 2.1: Conditions for comparing the two zero-frequency noise derivations. They describe the same physical bias and coincide after the dressed result is expanded to the same tunnelling order.

2.4.5 Thermal and Poisson checks of the shared limit

For equal temperatures and effectively unbounded flat bands, let $\Delta = \Delta\mu$ and $\beta = (k_BT)^{-1}$. Direct substitution of the Fermi functions gives, for $\Delta \neq 0$,

$$f_L(1-f_R) = \frac{f_L - f_R}{1 - e^{-\beta\Delta}}, \qquad f_R(1-f_L) = \frac{f_L - f_R}{e^{\beta\Delta} - 1}, \tag{2.4.36}$$

$$f_L(1-f_R) + f_R(1-f_L) = (f_L - f_R)\coth\frac{\beta\Delta}{2}. \tag{2.4.37}$$

Using $\int dE(f_L - f_R) = \Delta$ and constant $\mathcal{T}_0$ gives

$$\mathcal{S}_{\mathrm{tun}}^{\mathrm{sym}}(0) = \frac{e^2}{h}\mathcal{T}_0\Delta\coth\frac{\Delta}{2k_BT}, \qquad I_{\mathrm{tun}} = \frac{e}{h}\mathcal{T}_0\Delta. \tag{2.4.38}$$

Since $x\coth x \to 1$ as $x \to 0$, $\mathcal{S}_{\mathrm{tun}}^{\mathrm{sym}}(0) \to 2k_BT\mathcal{G}_{\mathrm{tun}}$, where $\mathcal{G}_{\mathrm{tun}} = e^2\mathcal{T}_0/h$. At zero temperature with nonzero fixed bias, $\Delta\coth(\Delta/2k_BT) \to |\Delta|$, so $\mathcal{S}_{\mathrm{tun}}^{\mathrm{sym}} = e|I_{\mathrm{tun}}|$. The doubled convention gives $4k_BT\mathcal{G}_{\mathrm{tun}}$ and $2e|I_{\mathrm{tun}}|$, respectively. These normalization checks agree with the explicit convention discussion in Ref. [4]. The noise is even under exchanging the reservoirs, whereas the current changes sign. These limits check both the factor of two and the physical meaning of the two occupation products.

Symbol table

The table summarizes the conventions used in the derivations. A reservoir index and a contour index refer to separate spaces.

Symbol	Meaning and domain
$\hbar, k_B, h, e$	$\hbar = k_B = 1$ unless restored; $h = 2\pi\hbar$; elementary charge $e > 0$, electron charge $-e$.
H, H_0	Physical real-time Hamiltonian and the declared disconnected/reference Hamiltonian.
$\mathcal{K}, \mathcal{K}_0$	Grand-canonical generators $H - \mu N$ and $\sum_\ell (H_\ell - \mu_\ell N_\ell)$.
$\rho_{\mathrm{eq}}, \rho_0$	Equilibrium density matrix and the specified initial density matrix.
$\ell = L, R$	Physical reservoirs; one channel per label in the displayed contact model.
$\kappa = +, -$	Forward/backward contour branches, ordered as $(+, -)$.
$c_i, c_{\ell k}, d$	Canonical mode annihilators; d belongs to the discrete dot.
$\psi_\ell(x)$	Continuum field $L_{\mathrm{sys}}^{-1/2} \sum_k e^{ikx} c_{\ell k}$, with dimension length$^{-1/2}$.
g, G	Reference/disconnected and full/coupled Green's functions.
$G^>, G^<$	$-i\langle c_i(t) c_j^\dagger(t')\rangle$ and $+i\langle c_j^\dagger(t') c_i(t)\rangle$; the first index belongs to the annihilator.
$G^T, G^{\bar{T}}$	Time-ordered and anti-time-ordered Green's functions.
G^R, G^A	Retarded and advanced single-fermion functions, built from the anticommutator and causal step functions.
G^K, G^M	Keldysh component $G^> + G^<$ and imaginary-time Matsubara function. For fermions, $G^K = -i\langle [c, c^\dagger]\rangle$.
$\Theta(t), 0^+$	Step function and positive analytic regulator; 0^+ is not a physical decay width.
ω, E, Ω	Internal frequency/energy, physical energy, and external measurement frequency. When units are restored $E = \hbar\omega$.
A, ν	Spectral matrix $i(G^R - G^A)$ and scalar local density of states $A_{ii}/(2\pi)$.
$f_\ell(E), f_0(\xi)$	Reservoir Fermi function $[e^{\beta(E-\mu_\ell)} + 1]^{-1}$ and its shifted-energy form.
n_{ij}	One-particle density matrix $\langle c_j^\dagger c_i\rangle = -iG_{ij}^<(t,t)$.
$\Sigma_d^R = \Lambda_d - i\Gamma_d$	Dot self-energy, real level shift, and nonnegative energy half-width; FWHM $= 2\Gamma_d$.
V_k, V_T	Discrete dot-lead hopping (energy) and continuum contact hopping (energy times length); both can be complex.
$\Delta\mu, V_{\mathrm{bias}}$	$\mu_L - \mu_R$ and the energy-bias voltage $\Delta\mu/e$.
λ_c	Dimensionless contact strength $\pi^2 \lvert V_T\rvert^2 \nu_L \nu_R$.
$\mathcal{T}, \mathcal{T}_0$	Dressed transmission $4\lambda_c/(1+\lambda_c)^2$ for flat leads, and its leading term $4\lambda_c$.
$\mathcal{G}$	Electrical conductance $\partial I/\partial V_{\mathrm{bias}}$; $e^2\mathcal{T}/h$ at zero temperature per channel.
$I, \delta I$	$-e\dot{N}_L$ and $I - \langle I\rangle$; positive I means electron number leaves L.
$C_{II}(t,t')$	Ordered connected current autocorrelation $\langle \delta I(t)\delta I(t')\rangle$.
$\mathcal{S}^{\mathrm{sym}}(\Omega)$	Two-sided symmetrized spectrum with prefactor $1/2$ in front of the current anticommutator.
$\mathcal{S}^{(2)}(\Omega)$	Twice $\mathcal{S}^{\mathrm{sym}}(\Omega)$; at zero frequency it is $2\int d\tau C_{II}(\tau)$.
$\mathsf{m}_C, \mathsf{L}_C$	Contour metric $\mathrm{diag}(1,-1)$ and explicit rotation to the triangular representation.
$\star, \circ_C$	Real-time and contour convolutions, including the ordered internal-index sums.
V, J	Hopping and current-vertex matrices in physical lead space.
$\widetilde{\psi}_\ell, \widetilde{g}_\ell$	Shifted-energy representation; $\widetilde{\psi}_\ell = e^{i\mu_\ell(t-t_0)}\psi_{\ell,H_0}$ and $\widetilde{g}_\ell(\omega) = g_{\ell,H_0}(\omega + \mu_\ell)$.

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